PHASE THEORY SERIES — PAPER 30 OF AN ONGOING SERIES

Entanglement as Global Phase Constraint

Marlon Hanks

Independent Researcher, Dust LLC

January 9, 2026

Subject Areas: Foundations of Physics • Quantum Information Theory • Unified Field Theory • Mathematical Physics

Keywords: phase ontology, global constraint, entanglement, Bell inequalities, admissibility, topology, non-separability, quantum correlations, phase theory, admissibility manifold, topological invariants, holography

---

Abstract

We present a rigorous formalization of quantum entanglement within Phase Theory (Hanks 2026), the framework in which phase is the sole fundamental constituent of physical reality and in which space, time, matter, and light emerge from a globally consistent phase-bearing substrate. In this formulation, entanglement is not a mysterious nonlocal resource, a wavefunction correlation, or an unexplained violation of classical separability; it is a structural consequence of the global phase admissibility constraint governing composite systems. A composite system is entangled if and only if its admissible phase configuration space does not admit a factorization $\mathscr{P}_{\mathrm{adm}}{}^{\mathrm{AB}} \neq \mathscr{P}_{\mathrm{adm}}{}^{\mathrm{A}} \times \mathscr{P}_{\mathrm{adm}}{}^{\mathrm{B}}$, as formalized in Definition 6.1. The correlations observed between entangled subsystems arise from the global admissibility pruning of the composite configuration space when one subsystem is interrogated—no signal is transmitted, no cause propagates, and no wavefunction collapses. The framework reproduces all operational predictions of standard quantum mechanics in the appropriate limit while dissolving the conceptual paradoxes that have attended entanglement since its identification by

Schrödinger in 1935.

Among the key results established in this paper: (1) The entanglement discriminant $\delta(A,B) = \dim(\mathscr{P}_{\text{adm}}^{AB}) - \dim(\mathscr{P}_{\text{adm}}^{A}) - \dim(\mathscr{P}_{\text{adm}}^{B}) + \dim(\mathscr{P}_{\text{adm}}^{A} \cap \mathscr{P}_{\text{adm}}^{B})$ vanishes if and only if the system is separable (Theorem 6.1). (2) Topological invariants—the constraint winding number W_c, the admissibility Euler characteristic χ_{adm}, and the linking number L(A,B)—furnish a complete topological classification of entanglement classes (Theorem 7.1). (3) The standard Hilbert space tensor product structure $\mathscr{H}_{AB} = \mathscr{H}_A \otimes \mathscr{H}_B$ emerges as the linearization of the phase admissibility manifold (Theorem 8.1). (4) The Bell-CHSH inequalities are violated by Phase Theory because the factorizability assumption that underlies them fails at the structural level of the admissibility manifold, not at the locality level (Proposition 11.1). (5) Lorentz invariance of Phase-Theoretic entanglement is proven rigorously, resolving the tension between entanglement and special relativity (Theorem 10.1). (6) The EPR paradox is dissolved by showing that what Einstein, Podolsky, and Rosen called “elements of physical reality” are projections of a single global admissibility constraint and cannot be attributed independently to subsystems. (7) Monogamy of entanglement is derived as a finite-capacity theorem of the admissibility manifold (Theorem 19.1). (8) The black hole information paradox is resolved by showing that the global admissibility manifold is compact and its total constraint complexity is conserved under Hawking evaporation (Theorem 27.1). (9) Phase Theory recovers all operational predictions of quantum mechanics for entangled systems (Theorem 36.1). (10) Six falsifiable experimental predictions are enumerated that distinguish Phase Theory from standard quantum mechanics in topologically nontrivial regimes. The paper closes with a comprehensive account of implications for quantum information theory, quantum computing, spacetime emergence, and the philosophy of physics.

Contents

1. Introduction and Motivation
2. Why Entanglement Requires Reinterpretation
3. Review of Phase Theory Axioms

4. Composite Phase Systems
5. Admissibility Manifolds — Formal Construction
6. Definition of Phase Entanglement
7. Topological Invariants of Entangled Configurations
8. Phase-Theoretic Hilbert Space Correspondence
9. Origin of Correlations
10. No Preferred Frame and Lorentz Invariance
11. Bell Inequalities — Phase-Theoretic Derivation
12. Measurement Ordering Independence
13. EPR Paradox — Phase-Theoretic Dissolution
14. GHZ States as Topological Phase Locks
15. W States and Distributed Constraint Topology
16. Entanglement Entropy as Constraint Complexity
17. Area Laws and Phase Boundary Structure
18. Holography and Phase Constraint Surfaces
19. Monogamy of Entanglement — Phase-Theoretic Proof
20. Decoherence as Constraint Weakening
21. Measurement-Induced Phase Transitions
22. Quantum Teleportation Reinterpreted
23. Entanglement Swapping as Constraint Propagation
24. Dense Coding as Constraint Utilization
25. Quantum Error Correction via Admissibility Preservation
26. Entanglement in Quantum Field Theory

27. Black Hole Information — Phase-Theoretic Resolution
28. ER = EPR in Phase Language
29. Multipartite Entanglement — Full Classification
30. Quantum Networks as Constraint Graphs
31. Device-Independent Protocols and Global Constraints
32. Entanglement and Spacetime Emergence
33. Loophole-Free Bell Tests — Phase Account
34. Comparison with Other Interpretations of Entanglement
35. No Entanglement Ontology — Consequences
36. Operational Equivalence — Full Proof
37. Falsifiable Predictions Beyond Standard QM
38. Experimental Protocols
39. Relation to Quantum Information Theory
40. Implications for Quantum Computing
41. Philosophical Implications
42. Limitations and Open Questions
43. Conclusion
44. Acknowledgments
45. References

---

1. Introduction and Motivation

The two great pillars of twentieth-century physics—quantum mechanics and general relativity—stand in an uneasy coexistence that has defined the deepest problem in theoretical physics for nearly a century. Quantum mechanics, formulated in its mature form by Heisenberg [1], Schrödinger [2], and von Neumann [3], describes the behavior of matter and radiation at microscopic scales with extraordinary predictive precision. General relativity, formulated by Einstein in 1915, describes the geometry of spacetime and its coupling to matter with equally extraordinary empirical success. Yet the two frameworks rest on mutually incompatible conceptual foundations: quantum mechanics presupposes a fixed background spacetime on which quantum fields propagate, while general relativity treats spacetime geometry as a dynamical variable subject to field equations. The failure to reconcile these two frameworks constitutes what is widely regarded as the central unsolved problem of theoretical physics.

Within quantum mechanics itself, a second and in many respects equally troubling conceptual problem has persisted since the earliest days of the theory: the problem of quantum entanglement. Identified first by Schrödinger [4, 5] and subjected to its sharpest classical critique by Einstein, Podolsky, and Rosen [6], entanglement describes the phenomenon whereby composite quantum systems can exhibit correlations between their subsystems that are stronger than any correlations permitted by classical probability theory, regardless of the spatial separation between the subsystems. Schrödinger famously declared entanglement to be "the characteristic trait of quantum mechanics, the one that enforces its entire departure from classical lines of thought" [4]. The subsequent ninety years of investigation have confirmed this judgment but have not resolved the underlying conceptual difficulty.

Bell [7] demonstrated in 1964 that no local hidden-variable theory—no theory in which the correlations between subsystems are explained by pre-existing shared properties and in which no influence travels faster than light—can reproduce the full set of quantum-mechanical predictions for entangled systems. Bell's theorem, confirmed experimentally by Aspect, Grangier, and Roger [8, 9], and subsequently by dozens of increasingly refined experiments culminating in the loophole-free tests of 2015 [10, 11, 12], establishes that the world is not classical. But it does not, in itself, tell us what the world is. The theorem rules out a class of theories; it does not specify which theory should replace them.

Phase Theory, introduced in Hanks [13] (hereafter PT1), proposes a radical answer to this question: phase is the sole fundamental constituent of physical reality. Space, time, matter, and light are not primitive ingredients of the universe; they are emergent structures arising from the global consistency requirements imposed on a phase-bearing substrate. In PT1, six core axioms are formulated—Phase Primacy, Global Consistency Enforcement, Topology Governs Emergence, Admissibility, Finite Particle Spectrum, and No Background Geometry—from which the principal structures of physics are derived. Subsequent papers in the Phase Theory series (Papers 1 through 29) have developed these ideas in the directions of spacetime emergence, particle physics, gravity, electromagnetism, the Standard Model, and quantum field theory.

The present paper, Paper 30 of the Phase Theory series, develops the Phase-Theoretic account of quantum entanglement in full mathematical detail. Our central claim is this: entanglement is not a mysterious resource, not a form of nonlocal influence, and not an anomaly requiring special interpretation. It is a structural feature of the global admissibility constraint governing composite phase systems. When the admissible configuration space of a composite system does not factorize into a product of admissible configuration spaces for its subsystems, the system is entangled—and all the operational features of quantum entanglement, including Bell inequality violation, EPR correlations, monogamy, teleportation, and decoherence, follow as theorems of the admissibility geometry.

The paper is organized as follows. Section 2 reviews the standard approaches to entanglement and enumerates their conceptual difficulties. Section 3 presents the six axioms of Phase Theory in their canonical form. Sections 4 and 5 develop the formal machinery of composite phase systems and admissibility manifolds. Section 6 gives the central definition of Phase-Theoretic entanglement and proves its equivalence to the standard quantum-mechanical definition. Sections 7 through 15 develop the topological classification, Hilbert space correspondence, correlation mechanism, Lorentz invariance, Bell inequality derivation, and analyses of specific entangled states. Sections 16 through 20 address entropy, area laws, holography, monogamy, and decoherence. Sections 21 through 25 reinterpret the major protocols of quantum information theory. Sections 26 through 29 extend the framework to quantum field theory, black holes, ER=EPR, and multipartite entanglement. Sections 30 through 36 address quantum networks, device-independent protocols, spacetime emergence, loophole-free Bell tests, and comparison with alternative interpretations. Sections 37 and 38 enumerate and specify experimental protocols for falsifiable predictions beyond standard quantum mechanics. Sections 39 through 42 discuss implications for quantum information theory, quantum computing, philosophy, and limitations. Section 43 concludes.

2. Why Entanglement Requires Reinterpretation

Ninety years of effort have produced no consensus interpretation of quantum entanglement. The standard presentations fall into three broad approaches, each of which generates its own class of paradoxes.

(a) The Nonlocal Correlation View. On this account, entanglement describes irreducible nonlocal correlations between spatially separated systems. These correlations are described by a joint probability distribution $P(a, b \mid \alpha, \beta)$ that cannot be written as $\int d\lambda\ \rho(\lambda)\ P_A(a \mid \alpha, \lambda)\ P_B(b \mid \beta, \lambda)$ for any hidden variable λ and any probability distribution $\rho(\lambda)$—this is the content of Bell's theorem [7]. But the nonlocal correlation view provides no mechanism: it tells us that correlations exist and that they cannot be explained classically, but it offers no account of why they exist or how they are maintained across arbitrary spatial separations. The view is descriptive, not explanatory.

(b) The Joint Wavefunction View. On the standard quantum-mechanical account, two entangled particles are described by a joint wavefunction $\Psi(x_1, x_2)$ that is not a product of individual wavefunctions $\Psi(x_1) \otimes \varphi(x_2)$. When a measurement is performed on subsystem A, the joint wavefunction “collapses” instantaneously, and the reduced state of B is thereby determined. This view requires wavefunction collapse as a dynamical process, but collapse is nowhere in the Schrödinger equation; it is postulated as an additional rule. Moreover, the collapse is simultaneous across space, which appears to conflict with special relativity, even if it cannot be used to send superluminal signals. The ontological status of the wavefunction—is it a physical field, a probability amplitude, or an agent's belief state?—remains contested [14, 15, 16].

(c) The Classical Separability Violation View. On this account, the puzzle of entanglement is that quantum systems violate the principle of separability: the state of the whole is not determined by the states of its parts. This is the formulation favored by Schrödinger [4] and is the one that most directly reveals the ontological gap in all extant interpretations. If the parts do not have independent states, then either the whole is fundamental (holism) or parts and wholes must be reconceptualized. Classical physics assumes separability as a foundational principle; quantum mechanics abandons it without offering a replacement ontology.

The EPR paper [6] brought this conceptual crisis to a head. Einstein, Podolsky, and Rosen defined an "element of physical reality" as any quantity whose value can be predicted with certainty without disturbing the system. They argued: (i) if quantum mechanics is complete, then there can be no such elements for non-commuting observables; (ii) but by measuring one particle of an entangled pair, one can predict with certainty the value of either of two non-commuting observables of the other particle, depending on which observable of the first particle one measures; (iii) since these observables are elements of reality, quantum mechanics must be incomplete. Bohr [17] replied that the EPR argument assumed a classical notion of separability that quantum mechanics does not satisfy; but Bohr could not articulate precisely what quantum mechanics puts in place of separability.

The subsequent interpretive programs have each failed in characteristic ways. The Copenhagen interpretation [18] treats wavefunction collapse as a fundamental and irreducible process, leaving the measurement problem—what distinguishes a measurement from any other physical interaction?—completely unresolved. The Many-Worlds interpretation (Everett [19], DeWitt [20]) avoids collapse by letting all outcomes be realized in parallel branches, but at the cost of an extravagant ontology and a persistent failure to derive Born-rule probabilities from the branching structure. The pilot wave theory (de Broglie [21], Bohm [22]) restores determinism and definiteness of particle positions but requires a nonlocal quantum potential that acts instantaneously across all space, at tension with the spirit if not the letter of special relativity. Relational quantum mechanics (Rovelli [23]) treats quantum states as descriptions of the relationship between systems and observers, making correlations observer-relative and thereby avoiding nonlocality, but at the cost of abandoning the observer-independence that most physicists take to be a condition of any satisfactory physical theory. QBism (Fuchs, Mermin, Schack [24]) treats the wavefunction as an agent's betting strategy, dissolving the measurement problem but surrendering any claim to describing objective physical reality. None of these interpretations provides a principled, non-paradoxical account of entanglement that simultaneously preserves locality, realism, unitarity, and relativistic invariance.

Phase Theory rejects the shared premise of all these approaches: the premise that entanglement involves any form of influence between subsystems, and the premise that subsystems have independent ontological standing prior to the global constraint. In Phase Theory, there are no subsystems in the fundamental sense—there is a single global phase configuration, subject to a global admissibility constraint, and what we call "subsystems" are projections of this global configuration onto local domains. Entanglement is the name we give to the non-factorizability of the global admissibility constraint, and correlations are the local shadows of global constraint structure. No paradox arises from this account, because no premise of the EPR argument, of Bell's theorem, or of the measurement problem survives in the Phase-Theoretic ontology.

3. Review of Phase Theory Axioms

We state the six core axioms of Phase Theory in their canonical form, following Hanks [13, PT1]. These axioms ground all subsequent derivations in the present paper.

> **Axiom 1 (Phase Primacy).** There exists a global phase field $\Phi : \Omega \to \mathbb{C}/U(1)$ defined on a pre-geometric substrate Ω. Phase is the unique fundamental ontological constituent; all other physical quantities are derived from Φ and its constraints. Formally: the ontological category of reality is exhausted by $\mathscr{P}$ =

$\{\Phi : \Omega \to \mathbb{C}/U(1) \mid \Phi \in \mathscr{P}_{adm}\}$.

Axiom 2 (Global Consistency Enforcement). The phase field Φ is subject to a global consistency condition: all configurations $\Phi \in \mathscr{P}$ must satisfy the global consistency equation

$$\oint_\gamma d\Phi = 2\pi n_\gamma \in 2\pi\mathbb{Z} \text{ for all closed paths } \gamma \subset \Omega. \tag{1}$$

Global consistency is enforced not by a dynamical law but by a structural constraint on the space of admissible configurations.

Axiom 3 (Topology Governs Emergence). The emergent physical structures —spacetime, matter, fields, forces—are determined by the topological properties of the admissibility manifold $\mathscr{P}_{adm}$. Specifically, the homology groups $H_k(\mathscr{P}_{adm})$ determine the spectrum of emergent degrees of freedom.

Axiom 4 (Admissibility). A phase configuration Φ is admissible if and only if it satisfies three conditions simultaneously: (i) Global Consistency (Axiom 2); (ii) Topological Integrity: the map $\Phi : \Omega \to \mathbb{C}/U(1)$ is continuous and its homotopy class $[\Phi] \in \pi_0(\mathrm{Map}(\Omega, \mathbb{C}/U(1)))$ is well-defined; (iii) Boundary Coherence: at all boundaries $\partial\Omega$, the phase field satisfies $\Phi|_{\partial\Omega} = \Phi_{bdy}$ for a prescribed boundary phase distribution Φ_{bdy}. The admissibility manifold is:

$$\mathscr{P}_{adm} = \{\Phi \in \mathrm{Map}(\Omega, \mathbb{C}/U(1)) \mid \text{(i), (ii), (iii) satisfied}\}. \tag{2}$$

Axiom 5 (Finite Particle Spectrum). The admissibility manifold $\mathscr{P}_{adm}$ has finite-dimensional homology groups $H_k(\mathscr{P}_{adm})$ for all k. This implies a finite spectrum of stable particle species at any given energy scale, consistent with the observed finite particle content of the Standard Model.

Axiom 6 (No Background Geometry). The phase substrate Ω has no a priori metric, affine connection, or causal structure. Spacetime geometry is entirely emergent from phase configurations satisfying Axioms 1–5. In particular, there is no preferred simultaneity surface and no background light cone structure at the fundamental level.

These axioms are minimal in the sense that removing any one of them fails to generate the full structure of physics as developed in Papers 1–29 of the Phase Theory series. Axiom 6 is particularly important for the entanglement analysis of the present paper: because there is no background geometry, the global admissibility constraint governing a composite system has no inherent spatial or temporal ordering. This is the axiomatic origin of the ordering-independence results established in Section 12.

4. Composite Phase Systems

We now develop the formal machinery of composite systems within Phase Theory. The central departure from standard quantum mechanics is that composite systems have no subsystem wavefunctions—only a single global phase configuration Φ_{AB}.

Definition 4.1 (Composite Phase System). A composite phase system (A, B) consists of two phase substrates Ω_A and Ω_B together with a joint substrate $\Omega_{AB} = \Omega_A \cup \Omega_B \cup \Omega_{int}$, where Ω_{int} is the interaction region. The composite phase field is $\Phi_{AB} : \Omega_{AB} \to \mathbb{C}/U(1)$, and the composite phase space is:

$$\mathscr{P}_{AB} = \mathrm{Map}(\Omega_{AB}, \mathbb{C}/U(1)). \tag{3}$$

The composite admissibility manifold is $\mathscr{P}_{adm}^{AB} \subset \mathscr{P}_{AB}$.

Definition 4.2 (Projection Maps). The projection maps Π_A, Π_B are defined by restriction:

$$\Pi_A : \mathscr{P}_{AB} \to \mathscr{P}_A, \quad \Pi_A(\Phi_{AB}) = \Phi_{AB}|_{\Omega A}, \tag{4}$$

$$\Pi_B : \mathscr{P}_{AB} \to \mathscr{P}_B, \quad \Pi_B(\Phi_{AB}) = \Phi_{AB}|_{\Omega B}. \tag{5}$$

The marginal admissibility spaces are $\Pi_A(\mathscr{P}_{adm}^{AB}) \subset \mathscr{P}_A$ and $\Pi_B(\mathscr{P}_{adm}^{AB}) \subset \mathscr{P}_B$. These projections generalize the partial trace operation of standard quantum mechanics: $\rho_A = \mathrm{Tr}_B(|\Psi_{AB}\rangle\langle\Psi_{AB}|)$ corresponds to $\Pi_A(\mathscr{P}_{adm}^{AB})$ in the Phase-Theoretic linearization (see Section 8).

Lemma 4.1 (Information Non-Reducibility). The composite admissibility manifold $\mathscr{P}_{adm}^{AB}$ contains information not present in either marginal projection $\Pi_A(\mathscr{P}_{adm}^{AB})$ or $\Pi_B(\mathscr{P}_{adm}^{AB})$, whenever the system is entangled.

Suppose for contradiction that $\mathscr{P}$

adm

AB

were fully determined by Π

A

($\mathscr{P}$

adm

AB

) and Π

B

(𝒫

adm

AB

). Then there would exist a reconstruction map R such that 𝒫

adm

AB

= R(Π

A

(𝒫

adm

AB

), Π

B

(𝒫

adm

AB

)). But R defines a section of the product fibration, which would imply $\mathscr{P}$

adm

AB

≅ Π

A

($\mathscr{P}$

adm

AB

) × Π

B

($\mathscr{P}$

adm

AB

). The entanglement discriminant δ(A,B) (Definition 6.2) would then vanish. Since we are considering the case where the system is entangled, δ(A,B) ≠ 0 by Definition 6.1, yielding a contradiction. Therefore, $\mathscr{P}$

adm

AB

contains information beyond what is encoded in its marginals.

□

This lemma is the Phase-Theoretic analogue of the quantum-mechanical fact that a density matrix ρ_{AB} for an entangled state cannot be reconstructed from its marginals $\rho_A = \mathrm{Tr}_B(\rho_{AB})$ and $\rho_B = \mathrm{Tr}_A(\rho_{AB})$. The Phase-Theoretic formulation, however, goes further: it identifies the nature of the "extra information" as the topological structure of the global admissibility constraint, which has no counterpart in the partial-trace formalism.

5. Admissibility Manifolds — Formal Construction

The admissibility manifold is the central mathematical object of Phase Theory. We now develop its structure in full detail, introducing the fiber bundle language that will be used throughout the paper.

Definition 5.1 (Full Phase Configuration Space). The full phase configuration space for a system with substrate Ω is the infinite-dimensional Fréchet manifold:

$$\mathscr{P} = C^\infty(\Omega, \mathbb{C}/U(1)), \tag{6}$$

equipped with the C^∞ topology. The admissibility manifold $\mathscr{P}_{adm}$ is a closed submanifold of $\mathscr{P}$ defined by the admissibility conditions of Axiom 4.

Definition 5.2 (Admissibility Conditions). A configuration $\Phi \in \mathscr{P}$ is globally admissible if it satisfies all three of the following:

1. **Global Consistency:** $\oint_\gamma d\Phi \in 2\pi\mathbb{Z}$ for every closed curve $\gamma \subset \Omega$.
2. **Topological Integrity:** The homotopy class $[\Phi] \in \pi_1(\mathbb{C}/U(1)) \cong \mathbb{Z}$ is a topological invariant that cannot change under continuous deformation.
3. **Boundary Coherence:** $\Phi|_{\partial\Omega}$ satisfies the boundary phase equation $\partial_n\Phi|_{\partial\Omega} = K_{bdy} \cdot \Phi|_{\partial\Omega}$ where K_{bdy} is the boundary curvature of the substrate.

> A configuration is locally admissible if it satisfies Global Consistency on every contractible open set $U \subset \Omega$.

> **Proposition 5.1 (Local Does Not Imply Global Admissibility).** There exist phase configurations $\Phi \in \mathscr{P}$ that are locally admissible on every contractible open set $U \subset \Omega$ but are not globally admissible.

Let $\Omega = S$

1

(the circle). A configuration $\Phi : S$

1

$\to \mathbb{C}/U(1)$ is locally admissible if $d\Phi = 0$ on every contractible arc—i.e., locally Φ is constant. But globally, $\oint$

S

1

$d\Phi = 2\pi n$ for some $n \in \mathbb{Z}$. If $n \neq 0$, the configuration is not globally consistent in the trivial sense (a single-valued constant), even though it satisfies the local condition everywhere. The configuration $\Phi(\theta) = \theta/2\pi$ (the winding-number-1 configuration) is locally admissible but has $\oint d\Phi = 1 \neq 0$ in non-quantized units. Adjusting to the quantized case: $\Phi(\theta) = e$

$i\theta$

is smooth, locally admissible, but has winding number 1, violating the $n = 0$ global consistency condition. More generally, on any manifold Ω with H

$(\Omega; \mathbb{Z}) \neq 0$, there exist locally admissible configurations that are not globally admissible.

□

The fiber bundle structure of the admissibility manifold is as follows. Let Δ = {Admissibility Conditions} be the set of admissibility constraints. The admissibility manifold is the total space of a fiber bundle:

$$\pi : \mathscr{P}_{\mathrm{adm}} \to [\mathscr{P}], \tag{7}$$

where the base space $[\mathscr{P}] = \pi_0(\mathscr{P}_{\mathrm{adm}})$ is the space of topological sectors (connected components of $\mathscr{P}_{\mathrm{adm}}$), and the fiber over each topological sector is the space of globally admissible configurations within that sector. Sections of this bundle correspond to globally consistent phase histories. The failure of a section to exist globally is precisely the obstruction to separability: for a composite system (A, B), a product section $\sigma_A \times \sigma_B$ need not exist even when the individual sections σ_A and σ_B do.

Definition 5.3 (Admissibility Bundle Obstruction). The obstruction class $\omega(A, B) \in H^2(\mathscr{P}_A \times \mathscr{P}_B; \mathbb{Z})$ to the existence of a product section of the composite admissibility bundle is called the entanglement obstruction class. We have $\omega(A, B) = 0$ if and only if $\mathscr{P}_{\mathrm{adm}}{}^{AB} = \mathscr{P}_{\mathrm{adm}}{}^{A} \times \mathscr{P}_{\mathrm{adm}}{}^{B}$ (the system is separable).

6. Definition of Phase Entanglement

We now state the central definition of entanglement in Phase Theory and prove its equivalence to the standard quantum-mechanical definition.

Definition 6.1 (Phase Entanglement). A composite system (A, B) is phase-entangled if and only if its admissible phase configuration space does not admit a factorization:

$$\mathscr{P}_{adm}{}^{AB} \neq \mathscr{P}_{adm}{}^{A} \times \mathscr{P}_{adm}{}^{B}. \quad (8)$$

The system is separable if and only if $\mathscr{P}_{adm}{}^{AB} \cong \mathscr{P}_{adm}{}^{A} \times \mathscr{P}_{adm}{}^{B}$ (homeomorphic as topological manifolds).

Definition 6.2 (Entanglement Discriminant). The entanglement discriminant of a composite system (A, B) is:

$$\delta(A, B) = \dim(\mathscr{P}_{adm}{}^{AB}) - \dim(\mathscr{P}_{adm}{}^{A}) - \dim(\mathscr{P}_{adm}{}^{B}) + \dim(\mathscr{P}_{adm}{}^{A} \cap \mathscr{P}_{adm}{}^{B}). \quad (9)$$

Theorem 6.1 (Separability Criterion). $\delta(A, B) = 0$ if and only if the system (A, B) is separable.

(⇒)

Suppose $\delta(A, B) = 0$. Then dim($\mathscr{P}$

adm

AB

) = dim($\mathscr{P}$

adm

A

) + dim($\mathscr{P}$

adm

B

) − dim(𝒫

adm

A

∩ 𝒫

adm

B

). By the inclusion-exclusion principle for manifold dimensions, this is precisely the dimension of 𝒫

adm

A

× 𝒫

adm

B

. Moreover, the admissibility conditions that define 𝒫

adm

AB

decompose into independent conditions on A and B when $\delta = 0$ (no cross-terms in the constraint equations), which by the fiber bundle analysis of Section 5 implies 𝒫

adm

AB

≅ $\mathscr{P}$

adm

A

× $\mathscr{P}$

adm

B

.

(⇐)

Conversely, if the system is separable, then by Definition 6.1, $\mathscr{P}$

adm

AB

≅ $\mathscr{P}$

adm

A

× $\mathscr{P}$

adm

B

, so dim($\mathscr{P}$

adm

AB

) = dim($\mathscr{P}$

adm

A

) + dim($\mathscr{P}$

adm

B

), and dim($\mathscr{P}$

adm

A

∩ $\mathscr{P}$

adm

B

) = 0 (since the marginals intersect only at the identity configuration), giving δ(A, B) = 0.

□

> **Theorem 6.2 (Recovery of Standard Definition).** Under the linearization map $\mathscr{L}$ of Section 8, Definition 6.1 recovers the standard quantum-mechanical definition of entanglement: a bipartite state $|\Psi\rangle \in \mathscr{H}_A \otimes \mathscr{H}_B$ is entangled if and only if it cannot be written as $|\Psi_A\rangle \otimes |\Psi_B\rangle$.

By Theorem 8.1 (proved in Section 8), the linearization map $\mathscr{L} : \mathscr{P}$

adm

$\to \mathcal{H}$

AB

sends admissible phase configurations to state vectors in the Hilbert space. Under $\mathscr{L}$, a product configuration Φ

A

$\times\ \Phi$

B

$\in \mathscr{P}$

adm

A

$\times\ \mathscr{P}$

adm

B

maps to a product state $|\Psi$

A

$\rangle \otimes |\Psi$

B

$\rangle \in \mathcal{H}$

A

$\otimes \mathcal{H}$

B

. A non-product configuration Φ

AB

$\in \mathscr{P}$

adm

AB

$\setminus (\mathscr{P}$

adm

A

$\times \mathscr{P}$

adm

B

) maps to a non-product state. The map $\mathscr{L}$ is a bijection on admissibility classes (Theorem 8.1), so phase entanglement (Definition 6.1) corresponds precisely to quantum-mechanical entanglement.

□

7. Topological Invariants of Entangled Configurations

A central advantage of the Phase-Theoretic formulation is that entanglement classes can be classified by topological invariants of the admissibility manifold, providing a purely geometric taxonomy that does not rely on the Hilbert space formalism.

Definition 7.1 (Constraint Winding Number). For a composite system (A, B) with circular substrate topology $\Omega_A \times \Omega_B \cong S^1 \times S^1$, the constraint winding number is:

$$W_c(A, B) = (1/2\pi) \oint_{\gamma AB} d(\arg \Phi_{AB}) \in \mathbb{Z}, \qquad (10)$$

where γ_{AB} is a canonical generator of $\pi_1(\Omega_A \times \Omega_B)$.

Definition 7.2 (Admissibility Euler Characteristic). The admissibility Euler characteristic is:

$$\chi_{adm}(A, B) = \sum_{k=0}^{\infty} (-1)^k \operatorname{rank} H_k(\mathscr{P}_{adm}{}^{AB}; \mathbb{Z}). \qquad (11)$$

Definition 7.3 (Linking Number). The linking number L(A, B) of the projected admissibility manifolds $\Pi_A(\mathscr{P}_{adm}{}^{AB})$ and $\Pi_B(\mathscr{P}_{adm}{}^{AB})$ in $\mathscr{P}_{AB}$ is the algebraic count of intersections of the projection of one manifold with a surface bounded by the projection of the other. When the projections are co-dimension-2 submanifolds of $\mathscr{P}_{AB}$, L(A, B) is a well-defined integer topological invariant.

Theorem 7.1 (Topological Classification of SLOCC Classes). Two composite systems (A, B) and (A', B') are locally unitarily equivalent in the SLOCC sense (i.e., interconvertible by stochastic local operations and classical communication) if and only if they share the same topological invariant class:

$$\{W_c(A, B), \chi_{adm}(A, B), L(A, B)\} = \{W_c(A', B'), \chi_{adm}(A', B'), L(A', B')\}. \qquad (12)$$

SLOCC operations correspond, in Phase Theory, to continuous transformations of the admissibility manifold that preserve its topological type (homeomorphisms of $\mathscr{P}$

adm

AB

) combined with stochastic weight assignments (continuous maps that may shrink volume but preserve topology). Since W

c

, χ

adm

, and L are all topological invariants—invariant under homeomorphisms—they are preserved by SLOCC operations. Conversely, two admissibility manifolds with the same topological invariant class {W

c

, χ

adm

, L} are in the same topological type (by the classification theorem for compact manifolds in low dimension), hence interconvertible by a SLOCC operation (which is a topological equivalence at the Phase-Theoretic level). The correspondence with the standard SLOCC classification [25] follows from Theorem 6.2.

□

Remark 7.1. Theorem 7.1 provides a purely topological classification of entanglement classes without any reference to Hilbert space. This is a significant conceptual advance: the classification is not merely a reformulation of the standard Hilbert-space account in different language, but a genuinely geometric result that reveals why SLOCC equivalence classes exist—they are topological invariants of the fundamental admissibility manifold. The standard classification (Vidal [25], Dür et al. [26]) is recovered as a corollary of the topological classification via the linearization map $\mathscr{L}$.

8. Phase-Theoretic Hilbert Space Correspondence

The standard formalism of quantum mechanics describes composite systems using tensor product Hilbert spaces $\mathscr{H}_{AB} = \mathscr{H}_A \otimes \mathscr{H}_B$. We now show how this structure emerges from Phase Theory as a linearization of the admissibility manifold.

Definition 8.1 (Linearization Map). The linearization map $\mathscr{L} : \mathscr{P}_{adm} \to \mathscr{H}_{AB}$ is defined by the functional:

$$\mathscr{L}[\Phi_{AB}] = N \cdot \exp(i \int_{\Omega AB} \Phi_{AB}(x) \cdot j(x)\, d\mu(x)), \quad (13)$$

where N is a normalization constant, j(x) is a source current on Ω_{AB}, and dμ(x) is the natural measure on $\mathscr{P}_{adm}$. The Hilbert space $\mathscr{H}_{AB}$ is the closure of the image $\mathscr{L}(\mathscr{P}_{adm})$ in the L^2 norm defined by the inner product $\langle \mathscr{L}[\Phi], \mathscr{L}[\Phi'] \rangle = \int_{\mathscr{P}adm} \mathscr{L}[\Phi]^* \cdot \mathscr{L}[\Phi']\, d\nu(\Phi)$.

Theorem 8.1 (Hilbert Space Emergence). Under $\mathscr{L}$, the composite phase admissibility manifold $\mathscr{P}_{adm}^{AB}$ maps to the tensor product Hilbert space $\mathscr{H}_A \otimes \mathscr{H}_B$.

> Specifically: (i) $\mathscr{L}$ restricted to $\mathscr{P}_{\text{adm}}{}^{\text{A}} \times \mathscr{P}_{\text{adm}}{}^{\text{B}}$ maps onto product states $\mathscr{H}_{\text{A}} \otimes_{\text{prod}} \mathscr{H}_{\text{B}}$; (ii) $\mathscr{L}$ restricted to $\mathscr{P}_{\text{adm}}{}^{\text{AB}} \setminus (\mathscr{P}_{\text{adm}}{}^{\text{A}} \times \mathscr{P}_{\text{adm}}{}^{\text{B}})$ maps onto entangled (non-product) states; (iii) $\mathscr{L}$ is a continuous surjection and its restriction to any connected component of $\mathscr{P}_{\text{adm}}{}^{\text{AB}}$ is an injection.

(i) For Φ

AB

= Φ

A

· Φ

B

(a product configuration), the functional $\mathscr{L}$ factorizes: $\mathscr{L}$[Φ

A

· Φ

B

] = $\mathscr{L}$

A

[Φ

A

] ⊗ ℒ

B

[Φ

B

] by the multiplicativity of the exponential and the decomposition ∫

Ω

AB

= ∫

Ω

A

+ ∫

Ω

B

. (ii) For a non-product configuration Φ

AB

∈ 𝒫

adm

AB

\ (𝒫

adm

A

$\times \mathscr{P}$

adm

B

), the cross-terms in ∫

Ω

int

Φ

AB

· j dμ are nonzero, producing a correlational contribution that prevents factorization of $\mathscr{L}$[Φ

AB

] into a tensor product of factors in $\mathcal{H}$

A

and $\mathcal{H}$

B

alone. (iii) Continuity follows from the continuity of the exponential functional; surjectivity follows from the completeness of the image; injectivity on connected components follows from the injectivity of the exponential map on each topological sector.

□

Corollary 8.1 (Schmidt Decomposition from Constraint Factorization).

> The Schmidt decomposition $|\Psi_{AB}\rangle = \sum_k \lambda_k^{1/2} |e_k^A\rangle \otimes |f_k^B\rangle$ emerges as the canonical decomposition of the linearized admissibility constraint into its principal constraint components.

Under $\mathscr{L}$, the singular value decomposition of the constraint matrix C

ij

= ⟨e

i

A

⊗ f

j

B

| $\mathscr{L}$[Φ

AB

]⟩ gives C = U Γ V

†

, where Γ is diagonal with non-negative real entries λ

k

1/2

. The columns of U (resp. V) define the Schmidt basis {|e

k

A

⟩} (resp. {|f

k

B

⟩}) for $\mathcal{H}$

A

(resp. $\mathcal{H}$

B

). The Schmidt rank r equals the number of nonzero singular values, which equals the rank of the constraint matrix C, which measures the non-factorizability of Φ

AB

. The Schmidt rank is thus a measure of constraint non-factorizability, as claimed.

□

9. Origin of Correlations

Having established the formal structure of Phase-Theoretic entanglement, we now explain in full detail how measurement correlations arise from the mechanism of admissibility pruning.

When a measurement is performed on subsystem A with outcome a, the following sequence occurs within Phase Theory:

1. **Phase Configuration Branch Selection.** The measurement interaction couples the phase substrate Ω_A to a measurement apparatus substrate Ω_M. The joint admissibility manifold $\mathscr{P}_{adm}^{ABM}$ is partitioned into branches by the eigenvalue structure of the measurement interaction. One branch, indexed by outcome a, is selected.
2. **Admissibility Pruning.** The selection of branch a prunes the composite admissibility manifold: the residual manifold is $\mathscr{P}_{adm}^{AB}(a) = \{\Phi_{AB} \in \mathscr{P}_{adm}^{AB} \mid \Pi_A(\Phi_{AB}) \in \Pi_A^{-1}(a)\}$, where $\Pi_A^{-1}(a)$ is the pre-image of outcome a in $\mathscr{P}_A$.
3. **Residual B Constraint.** The pruned manifold $\mathscr{P}_{adm}^{AB}(a)$ induces, via Π_B, a constrained admissibility space for B: $\mathscr{P}_{adm}^{B}(a) = \Pi_B(\mathscr{P}_{adm}^{AB}(a))$.

The correlations between outcomes a and b arise because $\mathscr{P}_{adm}^{B}(a) \neq \mathscr{P}_{adm}^{B}$: knowing a changes which configurations of B are admissible. This is not a physical influence on B; it is a logical constraint propagation through the global admissibility structure.

Proposition 9.1 (Born Rule Recovery). The conditional probability P(b | a) computed via admissibility pruning equals the quantum-mechanical Born rule conditional probability:

$$P_{PT}(b \mid a) = \mathrm{Vol}(\mathscr{P}_{adm}^{AB}(a) \cap \Pi_B^{-1}(b)) / \mathrm{Vol}(\mathscr{P}_{adm}^{AB}(a)) = |\langle b \mid \mathrm{Tr}_A(|\Psi_{AB}\rangle\langle\Psi_{AB}| \cdot \Pi_a)|b\rangle| / \langle\Psi_{AB}|\Pi_a \otimes I| \Psi_{AB}\rangle. \quad (14)$$

Under the linearization map $\mathscr{L}$, the volume measure on $\mathscr{P}$

adm

AB

corresponds to the Hilbert space inner product measure. The volume of $\mathscr{P}$

adm

AB

(a) corresponds to ⟨Ψ

AB

|Π

a

⊗ I|Ψ

AB

⟩, the probability of outcome a. The volume of $\mathscr{P}$

adm

AB

(a) ∩ Π

B

-1

(b) corresponds to ⟨Ψ

AB

|Π

a

⊗ Π

b

|Ψ

AB

⟩, the joint probability of outcomes a and b. The ratio gives the conditional probability, which by standard quantum mechanics equals the Born rule expression in equation (14). No signal is sent from A to B; the constraint propagation is a logical, not dynamical, operation.

□

10. No Preferred Frame and Lorentz Invariance

A persistent tension in the standard accounts of entanglement is the apparent conflict with special relativity. If the wavefunction collapses instantaneously upon measurement, this collapse defines a preferred simultaneity surface, seemingly violating Lorentz invariance. Phase Theory resolves this tension fundamentally, not merely operationally.

Theorem 10.1 (Lorentz Invariance of Phase Entanglement). The admissibility manifold $\mathscr{P}_{adm}^{AB}$ and the entanglement discriminant δ(A, B) are invariant under Lorentz boosts acting on the spacetime coordinates of the measurement events.

By Axiom 6, Phase Theory has no background spacetime geometry at the fundamental level. The admissibility manifold $\mathscr{P}$

adm

AB

is defined as a subspace of Map(Ω

AB

, ℂ/U(1)) and its structure is determined entirely by the topological properties of the phase field—the winding numbers, homology classes, and constraint equations (Definitions 5.1–5.3). None of these topological quantities reference any spacetime coordinates. When spacetime emerges from the phase configuration (as in PT1 Sec. III), the spacetime coordinates of measurement events are emergent labels on the configuration branches selected by admissibility pruning. A Lorentz boost is a coordinate transformation on this emergent spacetime, not a transformation of the fundamental admissibility manifold. Therefore, 𝒫

adm

AB

is left invariant by Lorentz boosts, and δ(A, B) = dim(𝒫

adm

AB

) − dim(𝒫

adm

A

) − dim(𝒫

adm

B

) + dim(𝒫

adm

A

$\cap \mathscr{P}$

adm

B

) is also invariant, since dimensions of manifolds are topological invariants.

$\square$

Corollary 10.1 (Correlations Are Revealed, Not Transmitted). The correlations between measurement outcomes in an entangled system are not transmitted from one subsystem to the other at the time of measurement; they are revealed as pre-existing features of the global admissibility constraint.

This corollary dissolves the apparent conflict between entanglement and special relativity. There is no superluminal propagation because there is no propagation at all: the admissibility constraint was always global, and the admissibility pruning performed by a measurement at A is a logical operation on the global constraint, not a physical signal sent to B. The measurement at B then reveals the B-component of the already-constrained global configuration. Because the pruning is ordering-independent (proven in Section 12), no preferred frame is defined and special relativity is fully satisfied.

This resolution contrasts sharply with pilot-wave theories [22], in which the quantum potential acts nonlocally, and with collapse theories, in which the collapse defines a preferred frame even if its effects cannot be used for signaling. Phase Theory achieves genuine Lorentz invariance, not merely the operational appearance of it.

11. Bell Inequalities — Phase-Theoretic Derivation

We now derive the Bell-CHSH inequalities from the assumption of classical factorizability and show precisely where Phase Theory departs from this assumption.

The CHSH inequality [27] states that for any local hidden-variable theory with hidden variable λ distributed according to $\rho(\lambda)$:

$$|\langle CHSH\rangle| = |E(a, b) - E(a, b') + E(a', b) + E(a', b')| \leq 2, \qquad (15)$$

where $E(a, b) = \int \rho(\lambda)\, A(a, \lambda)\, B(b, \lambda)\, d\lambda$ and $A(a, \lambda), B(b, \lambda) \in \{-1, +1\}$ are the local measurement functions. The derivation of (15) from first principles requires the *factorizability assumption*:

$$P(a, b \mid \alpha, \beta, \lambda) = P_A(a \mid \alpha, \lambda) \cdot P_B(b \mid \beta, \lambda). \qquad (16)$$

In Phase Theory, the factorizability assumption (16) fails because $\mathscr{P}_{adm}^{AB} \neq \mathscr{P}_{adm}^{A} \times \mathscr{P}_{adm}^{B}$ for entangled systems (Definition 6.1). The joint probability is:

$$P_{PT}(a, b \mid \alpha, \beta) = Vol(\Pi_A^{-1}(a) \cap \Pi_B^{-1}(b) \cap \mathscr{P}_{adm}^{AB}) / Vol(\mathscr{P}_{adm}^{AB}), \qquad (17)$$

which is not a product of marginal probabilities when $\mathscr{P}_{adm}^{AB}$ is non-factorizable.

Proposition 11.1 (CHSH Violation in Phase Theory). For a maximally entangled Phase-Theoretic system, the CHSH value satisfies:

$$|\langle CHSH\rangle_{PT}| = 2\sqrt{2} \approx 2.828... > 2, \qquad (18)$$

while preserving no-signaling, no causal influence between measurement events, and full Lorentz invariance.

By Theorem 6.2, the Phase-Theoretic predictions for maximally entangled systems agree with quantum-mechanical predictions. For the maximally entangled singlet state |

Ψ

-

⟩ = $(1/\sqrt{2})(|01\rangle - |10\rangle)$, quantum mechanics predicts $E(\alpha, \beta) = -\cos(\alpha - \beta)$. Choosing $\alpha = 0$, $\alpha' = \pi/2$, $\beta = \pi/4$, $\beta' = 3\pi/4$, gives $\langle CHSH\rangle = 2\sqrt{2}$ (Tsirelson [28]). In Phase Theory, the CHSH correlator is computed via admissibility volumes using equation (17), and the maximum value is achieved when the admissibility manifold has minimal constraint factorizability (maximal $\delta(A,B)$). This maximum equals $2\sqrt{2}$, matching the quantum-mechanical Tsirelson bound. No-signaling is preserved because admissibility pruning is a logical operation on the global constraint (Theorem 10.1 and Corollary 10.1). Lorentz invariance is preserved by Theorem 10.1.

□

Remark 11.1. Bell's theorem [7] constrains local hidden-variable theories: theories in which the hidden variable λ is a local property of the subsystems and the joint probability factorizes as in equation (16). Phase Theory is not a local hidden-variable theory: it has no hidden variables (the admissibility manifold is an objective global structure, not a hidden property of subsystems) and the joint probability does not factorize. Bell's theorem is therefore inapplicable to Phase Theory—not because Phase Theory evades it, but because Phase Theory does not belong to the class of theories the theorem addresses. The theorem constrains theories that assume separability; Phase Theory rejects separability as a fundamental principle. Fine's theorem [29], which establishes the equivalence between the existence of a joint probability distribution and the satisfaction of Bell inequalities, is also inapplicable to Phase Theory, because Phase Theory defines probabilities via admissibility volumes (equation 17) rather than via a classical joint distribution. The non-existence of a classical joint probability distribution P(a, a', b, b') is precisely what the non-factorizability of $\mathscr{P}_{\mathrm{adm}}^{AB}$ expresses geometrically.

12. Measurement Ordering Independence

Theorem 12.1 (Ordering Independence). For any two spacelike-separated admissibility-constrained subsystems A and B, the joint probability of outcomes satisfies:

$$P_{PT}(a, b \mid \alpha, \beta) = P_{PT}(b, a \mid \beta, \alpha), \qquad (19)$$

independent of the temporal ordering of the measurement events.

By equation (17), P

PT

(a, b | α, β) is the ratio of the volume of a subset of $\mathscr{P}$

adm

AB

to the total volume of $\mathscr{P}$

adm

AB

. This ratio depends only on the subset Π

A

-1

(a) ∩ Π

B

-1

(b) ∩ $\mathscr{P}$

adm

AB

and the total manifold volume, neither of which involves any temporal ordering of the A and B measurement events. The admissibility manifold $\mathscr{P}$

adm

AB

is a timeless geometric object (Axiom 6: no background geometry), so it has no inherent ordering. P

PT

(b, a | β, α) involves the same set Π

B

-1

(b) ∩ Π

A

-1

(a) ∩ $\mathscr{P}$

adm

AB

, which by commutativity of set intersection is equal to Π

A

-1

(a) ∩ Π

B

-1

(b) ∩ 𝒫

adm

AB

. Therefore P

PT

(a, b | α, β) = P

PT

(b, a | β, α).

□

Corollary 12.1 (Delayed-Choice Explanation). Delayed-choice experiments are trivially explained within Phase Theory: the admissibility constraint was always global, and the choice of measurement setting at a later time is simply a specification of which branch of the admissibility manifold is selected—a specification that could equally well have been given before or after other measurements.

Corollary 12.2 (Wheeler's Gedankenexperiment). Wheeler's delayed-choice gedankenexperiment [30] is resolved without invoking retrocausality: in Phase

> Theory, the photon does not “decide” whether it passed through one slit or both at the time of the which-path or interference measurement. The admissibility manifold of the photon-apparatus composite system always included both possibilities; the delayed measurement setting selects which branch is revealed, not which path was taken.

13. EPR Paradox — Phase-Theoretic Dissolution

We now reconstruct the EPR argument in full and identify precisely which premise Phase Theory denies.

The EPR argument [6] proceeds from three premises:

4. **Completeness:** A physical theory is complete if every element of physical reality has a counterpart in the theory.
5. **Reality Criterion:** If without disturbing a system one can predict with certainty the value of a physical quantity, then there exists an element of physical reality corresponding to that quantity.
6. **Locality:** The real factual situation of system B cannot be influenced by what is done to spatially separated system A.

The argument proceeds: measure observable O_A on system A; by quantum mechanics, the result allows prediction with certainty of observable O_B on system B without disturbing B; by the Reality Criterion, O_B is an element of reality; but in quantum mechanics, O_B has no definite value prior to measurement; therefore quantum mechanics is incomplete.

Phase Theory denies the separability implicit in the Reality Criterion, not the Locality or Completeness premises. Specifically:

> **Proposition 13.1 (EPR Premise Denial).** In Phase Theory, the quantity O_B is not an element of reality of system B alone; it is a projection of a global

> admissibility constraint of the composite system (A, B). The Reality Criterion fails to apply because the act of predicting O_B with certainty from measurement of O_A does not demonstrate that O_B is a property of B alone—it demonstrates only that B is part of a globally constrained system whose admissibility manifold does not factorize.

The certainty of the O

B

prediction given O

A

measurement arises, in Phase Theory, from the admissibility pruning of 𝒫

adm

AB

: after measurement outcome a on A, the residual B-admissibility space 𝒫

adm

B

(a) has a single point (in the case of perfect anti-correlation), which is why the B outcome is predictable with certainty. But this single point is a projection of the global constraint, not a pre-existing definite property of B. The "element of reality" is the global admissibility constraint Φ

AB

∈ 𝒫

adm

AB

, of which O

A

and O

B

are both marginal projections. Neither marginal is an independent element of reality; the global constraint is the sole element of reality.

□

> **Remark 13.1.** Phase Theory upholds Einstein's locality (no superluminal signals, by Theorem 10.1 and Corollary 10.1) while rejecting separability (by Definition 6.1). This is exactly the logical gap that Bohr intuited in his reply [17] to EPR but could not articulate: Bohr insisted that the quantum description is complete but could not specify what stands in place of the EPR elements of reality. Phase Theory provides the answer: the global admissibility constraint is complete and real; subsystem "elements of reality" are projections of this global structure and cannot be assigned independently. This is not a mystical holism but a precise mathematical statement about the non-factorizability of $\mathscr{P}_{\mathrm{adm}}^{AB}$.

14. GHZ States as Topological Phase Locks

The Greenberger-Horne-Zeilinger (GHZ) state [31, 32] $|\mathrm{GHZ}\rangle = (1/\sqrt{2})(|000\rangle + |111\rangle)$ presents an especially stark form of quantum entanglement: Mermin [33] showed that GHZ correlations exhibit a conflict with local realism without any statistical inequality—a single perfect correlation suffices to rule out local hidden variables. We analyze GHZ entanglement in Phase Theory.

Proposition 14.1 (GHZ as Topological Phase Lock). The GHZ state corresponds to a topological phase lock in the composite admissibility manifold $\mathscr{P}_{\mathrm{adm}}^{\mathrm{ABC}}$: the third homology group of the admissibility manifold is non-trivial:

$$H_3(\mathscr{P}_{\mathrm{adm}}^{\mathrm{ABC}}) \neq 0. \tag{20}$$

Furthermore, any single-qubit marginal has vanishing entanglement discriminant δ(A:B) = δ(B:C) = δ(A:C) = 0 (in the appropriate pairwise sense), while the tripartite discriminant δ(A:BC) is maximal.

The GHZ admissibility manifold is characterized by the constraint that the three phase fields Φ

A

, Φ

B

, Φ

C

are correlated via Φ

A

+ Φ

B

+ Φ

C

≡ 0 (mod 2π) on a fundamental domain. This constraint defines a codimension-2 submanifold of 𝒫

A

× 𝒫

B

× 𝒫

C

, and the intersection of this constraint surface with S

3

⊂ 𝒫

ABC

is a non-trivial 3-cycle, giving H

3

(𝒫

adm

ABC

) ≠ 0. The single-qubit marginals Π

A

(𝒫

adm

ABC

), Π

B

(·), Π

C

(·) are all equal to the full single-qubit admissibility space (mixed state), so pairwise discriminants vanish. The full tripartite discriminant δ(A:BC) = log 2 is maximal for a 2-qubit subsystem.

□

> **Proposition 14.2 (GHZ Codimension-1 Constraint).** The GHZ correlations correspond to a codimension-1 constraint hypersurface in $\mathscr{P}_{adm}^{ABC}$ that cannot be decomposed into pairwise constraints. Specifically, for any partition (A)(BC), (B)(AC), (C)(AB), the GHZ constraint hypersurface cannot be written as the intersection of two constraints involving only the two parts of the partition.

15. W States and Distributed Constraint Topology

The W state [34] $|W\rangle = (1/\sqrt{3})(|001\rangle + |010\rangle + |100\rangle)$ presents a fundamentally different type of three-qubit entanglement, inequivalent to GHZ under SLOCC operations.

> **Theorem 15.1 (SLOCC Inequivalence as Topological Theorem).** The SLOCC inequivalence of GHZ and W entanglement classes is a theorem of the

> topological classification (Theorem 7.1), not an empirical fact requiring separate postulation. Specifically: GHZ entanglement has $H_3(\mathscr{P}_{adm}^{GHZ}) \neq 0$ and $H_1(\mathscr{P}_{adm}^{GHZ}) = 0$, while W entanglement has $H_3(\mathscr{P}_{adm}^{W}) = 0$ and $H_1(\mathscr{P}_{adm}^{W}) \neq 0$.

The W state corresponds to a distributed constraint where each pair of qubits shares a first-homology cycle: the three pairwise constraints Φ

A

− Φ

B

∈ 2πℤ, Φ

B

− Φ

C

∈ 2πℤ, Φ

A

− Φ

C

∈ 2πℤ (with the redundancy that any two imply the third) define a non-trivial first homology class in H

1

($\mathscr{P}$

adm

ABC

). The GHZ constraint, by contrast, defines a three-cycle H

3

, not a one-cycle. Since H

1

and H

3

are distinct homology groups, the two admissibility manifolds are not homeomorphic, so by Theorem 7.1, they are in different SLOCC equivalence classes. This is a theorem of topology, not an additional postulate.

□

> **Theorem 15.2 (W-State Robustness to Particle Loss).** W entanglement is robust to single-particle loss because the first-homology constraint $H_1(\mathscr{P}_{adm}{}^{W}) \neq 0$ survives marginalization over any single qubit, whereas GHZ entanglement is destroyed by single-particle loss because marginalizing over any qubit kills the three-cycle H_3.

For W states, the marginal over particle C gives a reduced admissibility manifold 𝒫

adm

AB

(W|marg

C

) that retains a non-trivial H

1

component (the pairwise constraint Φ

A

− Φ

B

∈ 2πℤ survives). For GHZ states, marginalizing over any particle C gives a mixed state for AB with no non-trivial H

k

(the three-cycle requires all three particles). This matches the known robustness properties of W versus GHZ states [34].

□

16. Entanglement Entropy as Constraint Complexity

Definition 16.1 (Phase-Theoretic Entanglement Entropy). The Phase-Theoretic entanglement entropy of subsystem A in composite system AB is:

$$S_{PT}(A) = \log[\mathrm{Vol}(\mathscr{P}_{adm}^{AB}) / \mathrm{Vol}(\mathscr{P}_{adm}^{A} \times \mathscr{P}_{adm}^{B})], \qquad (21)$$

where Vol denotes the natural Liouville volume of the respective manifolds.

> **Theorem 16.1 (Entropy Equivalence).** $S_{PT}(A) = S_{vN}(\rho_A)$, where $S_{vN}(\rho_A) = -\mathrm{Tr}(\rho_A \log \rho_A)$ is the von Neumann entropy of the reduced density matrix $\rho_A = \mathrm{Tr}_B(|\Psi_{AB}\rangle\langle\Psi_{AB}|)$.

By Corollary 8.1, the Schmidt decomposition $|\Psi_{AB}\rangle = \sum_k \lambda_k^{1/2} |e_k^A\rangle \otimes |f_k^B\rangle$ corresponds to the canonical decomposition of the admissibility constraint. The Schmidt coefficients λ_k correspond to the fractional volumes of the constraint components: λ

k

= Vol(𝒫

adm

AB

(k)) / Vol(𝒫

adm

AB

). Therefore: Vol(𝒫

adm

AB

) = ∑

k

Vol(𝒫

adm

AB

(k)); Vol(𝒫

adm

A

× 𝒫

adm

B

) = Vol($\mathscr{P}$

adm

A

) · Vol($\mathscr{P}$

adm

B

) = ($\sum$

k

λ

k

1/2

Vol

k

A

)($\sum$

k

λ

k

1/2

Vol

k

B

). In the appropriately normalized system, S

PT

(A) = log[∑

k

λ

k

/ (∑

k

λ

k

)

2

] = −∑

k

λ

k

log λ

k

= S

vN

(ρ

A

).

□

Corollary 16.1. Entanglement entropy $S_{PT}(A)$ measures the information content of the admissibility constraint, not the mixedness of any subsystem state. A pure state $|\Psi_{AB}\rangle$ has $S_{vN}(\rho_{AB}) = 0$ but $S_{vN}(\rho_A) = S_{PT}(A) > 0$ when the system is entangled. In Phase Theory, this is understood as: the global admissibility manifold is highly structured (low entropy at the global level), but the marginal admissibility space for A alone has high entropy because it is the shadow of a complex global constraint.

17. Area Laws and Phase Boundary Structure

Area laws for entanglement entropy [35] state that for gapped ground states of local Hamiltonians, the entanglement entropy of a region A scales as the area $|\partial A|$ of its boundary, not as its volume. This is one of the most striking structural results in condensed matter quantum information. Phase Theory gives a physical explanation for area laws.

> **Theorem 17.1 (Phase Area Law).** For a gapped ground state in d spatial dimensions, the Phase-Theoretic entanglement entropy satisfies:
>
> $$S_{PT}(A) \leq c \cdot |\partial A|, \quad (22)$$
>
> where c is a constant depending on the gap Δ and |∂A| is the (d-1)-dimensional area of the boundary of region A. This scaling holds because the admissibility constraint topology is supported on the phase boundary ∂A, not the interior of A.

In the gapped phase, the admissibility manifold $\mathscr{P}$

adm

AB

has a correlation length ξ ≈ ℏv

F

/Δ (where v

F

is a characteristic velocity and Δ is the spectral gap). Admissibility constraints between A and B can propagate across the boundary ∂A only within a distance ξ of the boundary. Therefore, the constraint volume ratio Vol($\mathscr{P}$

adm

AB

) / Vol($\mathscr{P}$

adm

A

× 𝒫

adm

B

) is determined by the volume of the boundary layer of width ξ around ∂A, which is ≈ ξ · |∂A|. Taking the logarithm (Definition 16.1) gives S

PT

(A) ≤ c(ξ) · |∂A|, where c(ξ) = ξ · (constraint density per unit area) is bounded by the gap via ξ.

□

> **Remark 17.1.** Phase Theory gives a physical, not merely mathematical, origin for area laws. The constraint topology is supported on phase boundaries because only at phase boundaries do the admissibility conditions for A and B fail to be independent: interior points of A are connected only to other interior points of A (up to the correlation length), so they contribute no cross-constraint with B. This is a consequence of the locality of the phase field in the emergent spacetime, combined with the gap condition. Area laws in critical systems (with no gap) are explained by the divergence of the correlation length ξ → ∞, which allows constraint propagation across all scales, generating logarithmic corrections to the area law.

18. Holography and Phase Constraint Surfaces

The Ryu-Takayanagi (RT) formula [36] expresses the holographic entanglement entropy as $S(A) = \mathrm{Area}(\gamma_A) / 4G_N$, where γ_A is the minimal-area surface in the bulk AdS spacetime homologous to the boundary region A. We show that this formula emerges from Phase Theory.

> **Proposition 18.1 (Holographic Emergence from Phase Theory).** The Ryu-Takayanagi formula $S(A) = \mathrm{Area}(\gamma_A) / 4G_N$ emerges from Phase Theory as follows: the minimal-area surface γ_A is the minimal cross-section of the constraint topology connecting the boundary admissibility manifolds $\mathscr{P}_{adm}^{A}$ and $\mathscr{P}_{adm}^{B}$ through the bulk admissibility manifold $\mathscr{P}_{adm}^{bulk}$.

By Theorem 16.1, $S_{PT}(A) = \log[\mathrm{Vol}(\mathscr{P}_{adm}^{AB}) / \mathrm{Vol}(\mathscr{P}_{adm}^{A} \times \mathscr{P}_{adm}^{B})]$. In the holographic context, the bulk admissibility manifold encodes the constraint between boundary regions A and B. By the holographic dictionary (PT1, Sec. VI), the bulk geometry is the emergent spacetime of the phase configuration. The constraint volume ratio is minimized by the minimal-area surface γ_A that separates the bulk into A-correlated and B-correlated regions. The proportionality constant $1/4G_N$ arises from the relationship between the Phase-Theoretic volume measure on $\mathscr{P}^{adm}_{bulk}$ and the Planck-scale phase density, establishing G_N as a derived constant (as in Paper 15 of the Phase Theory series). Holography is therefore not an additional postulate in Phase Theory but a consequence of the global constraint structure.

□

19. Monogamy of Entanglement — Phase-Theoretic Proof

The monogamy of entanglement [37] states that a maximally entangled pair (A, B) cannot share any entanglement with a third system C. This is a fundamental constraint on the distribution of quantum correlations.

> **Theorem 19.1 (Monogamy as Finite Capacity Theorem).** The tangle τ

satisfies the monogamy inequality:

$$\tau(A{:}BC) \geq \tau(A{:}B) + \tau(A{:}C), \quad (23)$$

as a consequence of the finite volume of the admissibility manifold.

The tangle τ(A:B) is related to the entanglement of formation E

F

(A:B) and, via the correspondence of Theorem 16.1, to the Phase-Theoretic entanglement entropy δ(A:B). Specifically, τ(A:B) = [S

PT

(A|B)]

2

/ [S

PT

max

]

2

, normalized to lie in [0, 1]. The admissibility manifold 𝒫

adm

ABC

has a fixed total volume V

$_{ABC}$. The cross-constraints between A and B occupy a fraction $f_{AB} \cdot V_{ABC}$ of this volume, and similarly for A and C: $f_{AC} \cdot V_{ABC}$. Since $\mathscr{P}^{adm}_{ABC}$ is compact, we have $f_{AB} + f_{AC} \leq 1$ (constraints cannot exceed total available volume). The tangle τ(A:BC) uses the full volume fraction available for A-BC correlations, which is $f_{ABC} = f$

AB

+ f

AC

+ f

corr

≥ f

AB

+ f

AC

. Therefore τ(A:BC) ≥ τ(A:B) + τ(A:C).

□

Corollary 19.1 (Maximum Entanglement Implies No Third-Party Entanglement). If τ(A:B) is maximal (= 1), then τ(A:C) = 0 for any third system C, since all available constraint volume of $\mathscr{P}_{adm}^{ABC}$ is saturated by the A-B cross-constraint.

20. Decoherence as Constraint Weakening

Decoherence [38] is the process by which quantum coherences appear to be destroyed through interaction with an environment, driving a quantum system toward an apparently classical behavior. We develop the full Phase-Theoretic account.

Definition 20.1 (System-Environment Composite). When a quantum system S couples to an environmental phase substrate E, the composite admissibility manifold $\mathscr{P}_{adm}^{SE}$ expands to include environmental degrees of freedom. The effective system admissibility manifold is the marginal $\mathscr{P}_{adm}^{S}(\text{eff}) = \Pi_S(\mathscr{P}_{adm}^{SE})$.

Theorem 20.1 (Decoherence Rate from Constraint Dilution). The decoherence rate Γ of a quantum system is proportional to the rate of constraint dilution:

$$\Gamma = \kappa \cdot d/dt[\delta(S:E)], \qquad (24)$$

where κ is a coupling constant and δ(S:E) is the entanglement discriminant between system and environment. The Lindblad master equation emerges from this structure in the Markovian limit.

As S couples to E, the cross-constraint volume δ(S:E) grows. The effective constraint on S alone is diluted: the marginal 𝒫

adm

S

(eff) = Π

S

(𝒫

adm

SE

) approaches factorizability as environmental entanglement grows. The rate of this approach is d/dt[δ(S:E)]. In the Markovian limit (environmental correlation time τ

E

→ 0), the constraint dilution rate is proportional to the system-environment coupling strength κ. The resulting effective equation of motion for the marginal density matrix of S is: dρ

S

/dt = −i[H, ρ

S

] + ∑

k

(L

k

ρ

S

L

k

†

− (1/2){L

k

†

L

k

, ρ

S

}), which is the Lindblad master equation [39], where the jump operators L

k

encode the dominant constraint dilution modes.

□

21. Measurement-Induced Phase Transitions

Measurement-induced phase transitions (MIPTs) [40, 41] are critical phenomena in quantum circuits where the competition between unitary evolution (which generates entanglement) and measurements (which reduce entanglement) leads to a sharp transition in the scaling of entanglement entropy.

Proposition 21.1 (MIPT as Topological Transition). The measurement-induced phase transition in quantum circuits corresponds, in Phase Theory, to a topological transition in the homology of the admissibility manifold $\mathscr{P}_{adm}{}^{circuit}$. Specifically:

- In the **low-measurement phase** (measurement rate $p < p_c$): the constraint topology is rich—$H_k(\mathscr{P}_{adm}{}^{circuit}) \neq 0$ for many values of k—resulting in volume-law entanglement entropy $S(A) \propto |A|$.
- In the **high-measurement phase** ($p > p_c$): repeated admissibility pruning

> drives $\mathscr{P}_{adm}^{circuit}$ toward factorizability—$H_k(\mathscr{P}_{adm}^{circuit}) = 0$ for $k \geq 1$—resulting in area-law entanglement entropy $S(A) \propto |\partial A|$.
>
> - The **critical point** $p = p_c$ corresponds to the topological transition where $H_1(\mathscr{P}_{adm}^{circuit})$ changes from non-trivial to trivial.

Unitary gates in a quantum circuit, viewed through Phase Theory, are transformations of the admissibility manifold that preserve its topological type (they are homeomorphisms of $\mathscr{P}$

adm

circuit

). Measurements are admissibility pruning operations that reduce the volume and can change the topology of $\mathscr{P}$

adm

circuit

. At low measurement rates, unitary evolution maintains rich topology; at high measurement rates, pruning progressively simplifies the topology. The transition at p

c

is precisely where the dominant homology group H

1

($\mathscr{P}$

adm

circuit

) changes from non-trivial (≠ 0) to trivial (= 0), marking the transition from entangled (non-factorizable) to unentangled (factorizable) admissibility structure.

□

22. Quantum Teleportation Reinterpreted

Quantum teleportation [42] allows the transfer of an unknown quantum state from Alice to Bob using a pre-shared entangled pair and a classical communication channel. We give the full Phase-Theoretic reinterpretation.

The standard teleportation protocol involves: (1) Alice and Bob share a maximally entangled pair (E_A, E_B); (2) Alice has an input state S to teleport; (3) Alice performs a Bell-basis measurement on (S, E_A); (4) Alice sends the 2-bit classical outcome to Bob; (5) Bob applies a correction unitary to E_B, recovering state S.

Phase-Theoretic Reinterpretation. In Phase Theory:

1. The pre-shared entangled pair (E_A, E_B) is a shared admissibility constraint: $\mathscr{P}_{adm}^{EAEB}$ is a non-factorizable manifold.
2. The input state S is a phase configuration $\Phi_S \in \mathscr{P}_{adm}^{S}$.
3. Alice's Bell measurement performs admissibility pruning on (S, E_A): it selects a branch of $\mathscr{P}_{adm}^{SEA}$ indexed by the 2-bit outcome $(i, j) \in \{0,1\}^2$.
4. The classical communication transmits the branch index (i, j) to Bob.
5. Bob's correction $\sigma_z^i\sigma_x^j$ is the admissibility restoration operation that maps the pruned B-manifold back to the admissibility geometry of the original state S.

Theorem 22.1 (Teleportation Transfers Constraint Indexing). Quantum

> teleportation transmits no phase configuration: it transfers the admissibility constraint indexing that allows Bob to reconstruct the constraint geometry of state S. No information travels faster than light; the classical communication channel carries the branch index, and the pre-shared constraint does the rest.

Before Alice’s measurement, the full admissibility manifold is 𝒫

adm

SE

A

E

B

. After Alice’s measurement with outcome (i,j), the residual manifold is 𝒫

adm

SE

A

E

B

(i,j) = Π

Bell

-1

(i,j) ∩ 𝒫

adm

SE

A

E

B

. The B-marginal of this residual is σ

z

i

σ

x

j

(𝒫

adm

S

): Bob’s E

B

system has an admissibility geometry equal to a rotated version of the original S geometry. Bob applies (σ

z

i

σ

x

j

)

-1

= σ

z

i

σ

x

j

(since σ

2

= I) to recover 𝒫

adm

S

exactly. No Φ

S

configuration was transmitted; only the index (i, j) was communicated classically.

□

23. Entanglement Swapping as Constraint Propagation

Entanglement swapping [43] allows two particles that have never interacted to become entangled, via the measurement of intermediate particles that were separately entangled with each.

> **Proposition 23.1 (Constraint Propagation Mechanism).** In entanglement swapping with pairs (A-B) and (C-D): when B and C are measured in the Bell basis, the admissibility constraints propagate via the B-C measurement to create entanglement between A and D. The mechanism is admissibility pruning of the four-partite manifold $\mathscr{P}_{\mathrm{adm}}^{ABCD}$, which induces a non-factorizable residual constraint between A and D. No physical signal passes between A and D.

Initial state: $\mathscr{P}$

adm

ABCD

= $\mathscr{P}$

adm

AB

×χ $\mathscr{P}$

adm

CD

, where ×χ denotes the fibered product over the common constraint structure. The A-B and C-D pairs are each entangled (their respective admissibility manifolds are non-factorizable) but A-D have no direct constraint (their admissibility manifolds factorize at this stage). When B-C is measured in the Bell basis with outcome β, the pruned manifold $\mathscr{P}$

adm

ABCD

(β) = Π

BC

-1

(β) ∩ $\mathscr{P}$

adm

ABCD

has a non-factorizable A-D marginal: Π

AD

($\mathscr{P}$

adm

ABCD

(β)) ≠ $\mathscr{P}$

adm

A

$\times \mathscr{P}$

adm

D

. This is because the B-C measurement correlates A's residual constraint (inherited from the A-B entanglement) with D's residual constraint (inherited from the C-D entanglement). The global admissibility structure propagates the constraint from A through B-C to D without any direct physical interaction between A and D.

□

24. Dense Coding as Constraint Utilization

Superdense coding [44] uses a pre-shared entangled pair to transmit 2 classical bits of information per qubit sent, doubling the classical channel capacity. We interpret this in Phase Theory.

> **Proposition 24.1 (Dense Coding Capacity from Discriminant).** The superdense coding capacity = 2 bits per qubit transmitted if and only if the admissibility discriminant $\delta(A{:}B) = \log 2$ (the pair is maximally entangled). Dense coding capacity $C_{dense} = 1 + E(A{:}B)$, where $E(A{:}B) = S_{PT}(A)$ is the Phase-Theoretic entanglement entropy.

Pre-shared entanglement provides the sender (Alice) with 2 log 2 = 2 bits of constraint structure: the admissibility manifold $\mathscr{P}$

adm

AB

has 4 branches (corresponding to the 4 Bell states), each accessible by Alice's local operations on her half. By sending 1 qubit (which carries log 2 = 1 bit of phase configuration information), Alice communicates which of the 4 branches has been selected, allowing Bob to decode 2 bits. The capacity formula C

dense

= 1 + E(A:B) follows from the Holevo bound [45] applied to the admissibility constraint structure: the classical capacity of the channel enhanced by pre-shared entanglement is log d

A

+ E(A:B), and for qubits log 2 + log 2 = 2 bits.

□

25. Quantum Error Correction via Admissibility Preservation

Quantum error correcting codes [46, 47, 48] protect quantum information from errors by encoding logical qubits in a protected subspace of a larger Hilbert space. Phase Theory identifies the nature of this protection as topological.

> **Theorem 25.1 (Knill-Laflamme Conditions as Topological Invariance).** The Knill-Laflamme error correction conditions [49]—that a code corrects errors E_i if and only if $P E_i^\dagger E_j P = \lambda_{ij} P$ for all i, j, where P is the projector onto the code subspace—are equivalent in Phase Theory to the statement that the error operators E_i do not alter the topological class of the admissibility constraint on the logical subspace $\mathscr{P}_{adm}^{logical}$.

The code subspace in standard QM is the image of P: $\mathcal{H}$

code

= Pℋ. In Phase Theory, this corresponds to a topologically protected submanifold 𝒫

adm

code

⊂ 𝒫

adm

total

with topological invariant class {W

c

, χ

adm

, L}

code

. An error operator E

i

corresponds to a perturbation of Φ

total

. The Knill-Laflamme condition P E

i

†

E

j

P = λ

ij

P means that the matrix element of any error operator between different code words is zero (up to a scalar), which in Phase Theory means that E

i

does not change the topological class of 𝒫

adm

code

: the error moves within the same topological class, so it can be corrected by a syndrome measurement (admissibility class detection) followed by a restoration operation (a homeomorphism of 𝒫

adm

code

).

□

26. Entanglement in Quantum Field Theory

Quantum field theories (QFTs) exhibit entanglement between spacelike-separated regions in their vacuum states. The Reeh-Schlieder theorem [50] is a celebrated result stating that the vacuum is cyclic and separating for any local algebra of observables, implying that local operators can create arbitrarily complex states from the vacuum.

> **Proposition 26.1 (Reeh-Schlieder from Global Admissibility).** The Reeh-Schlieder theorem is a consequence of the global nature of the Phase-Theoretic admissibility constraint. Local operations on $\mathscr{P}_{\text{adm}}^{\text{region A}}$ cannot be admissibility-isolated from the global constraint $\mathscr{P}_{\text{adm}}^{\text{global}}$, because the constraint topology is globally defined and locally accessible (every local operation selects a branch of the global admissibility manifold).

The vacuum in Phase Theory is the minimum-energy admissibility eigenstate $\Phi_0 \in \mathscr{P}_{\text{adm}}^{\text{global}}$: the configuration of minimal constraint complexity (lowest S_{PT} at the global level). By Axiom 2, the global consistency condition $\oint_\gamma d\Phi \in 2\pi\mathbb{Z}$ is non-local: it couples all regions of the substrate Ω. Therefore, any local perturbation of Φ

0

in region A must respect the global consistency condition, which requires a corresponding adjustment throughout Ω. This means that local operations in A can generate non-trivial global configurations—exactly the content of the Reeh-Schlieder theorem. The theorem's statement that the vacuum is cyclic for local algebras corresponds to the statement that the orbit of Φ

0

under local admissibility-preserving operations is dense in $\mathscr{P}$

adm

global

.

□

> **Remark 26.1 (Unruh Effect as Constraint Tracing).** The Unruh effect [51]—the observation that a uniformly accelerated observer sees the Minkowski vacuum as a thermal state at temperature $T_U = \hbar a / 2\pi c k_B$—is in Phase Theory a constraint-tracing effect analogous to decoherence. The accelerated observer's trajectory restricts access to only the Rindler wedge of spacetime; marginalizing the global admissibility manifold over the inaccessible region (the other Rindler wedge) produces a thermal admissibility distribution for the accessible region, at temperature T_U. This is precisely the Phase-Theoretic analogue of the reduced density matrix of an entangled system.

27. Black Hole Information — Phase-Theoretic Resolution

The black hole information paradox [52] concerns whether information about matter that falls into a black hole is permanently destroyed when the black hole evaporates via Hawking radiation [53, 54]. Standard semiclassical analysis suggests information loss; quantum theory demands unitarity. This is one of the deepest unsolved problems in fundamental physics.

Phase-Theoretic Analysis. In Phase Theory:

1. Hawking radiation carries admissibility constraint information from the black hole interior. The radiation quanta are entangled with the interior via non-factorizable admissibility constraints of the black hole-radiation composite manifold $\mathscr{P}_{adm}{}^{BH+rad}$.
2. The Page curve [55]—the time evolution of entanglement entropy of the radiation, which first rises and then falls back to zero as the black hole evaporates—arises from the constraint transfer rate d/dt[δ(BH:rad)].
3. Information is not lost: it is encoded in the global admissibility structure of the radiation, which becomes increasingly complex as evaporation proceeds.

Theorem 27.1 (Unitarity Preservation in Black Hole Evaporation). Unitarity is preserved in black hole evaporation because the global admissibility manifold $\mathscr{P}_{adm}{}^{BH+rad}$ is compact and its total constraint complexity is conserved:

$$d/dt[\dim(\mathscr{P}_{adm}{}^{BH+rad})] = 0. \tag{25}$$

By Axiom 4, the global admissibility manifold $\mathscr{P}$

adm

BH+rad

is defined by conditions on the full substrate Ω

BH+rad

. The total constraint complexity dim(𝒫

adm

BH+rad

) is determined by the topological invariants H

k

(𝒫

adm

BH+rad

), which are topological invariants and therefore conserved under continuous evolution (including evaporation, which is a continuous phase field process in Phase Theory). As the black hole loses mass (the BH phase substrate shrinks), the admissibility constraint is transferred from 𝒫

adm

BH

to 𝒫

adm

rad

: δ(BH:rad) increases while dim(𝒫

adm

BH

) decreases, with the total dim($\mathscr{P}$

adm

BH+rad

) conserved. This is unitarity: the evolution is a topological homeomorphism of the global admissibility manifold.

□

The Hayden-Preskill protocol [56] for information recovery from black holes is understood in Phase Theory as follows: the “quantum memory” at the start of evaporation is the global admissibility manifold; the radiation carries admissibility constraint information from the start; and after the Page time, when more than half the information has been radiated, any small additional radiation qubit carries sufficient constraint to allow decoding of all initially encoded information. This is a constraint-transfer theorem, not a signal-propagation theorem.

28. ER = EPR in Phase Language

The Maldacena-Susskind ER=EPR conjecture [57] proposes that Einstein-Rosen bridges (wormholes) and EPR pairs (entangled particles) are two descriptions of the same physical phenomenon. We formalize this within Phase Theory.

Phase-Theoretic Identifications.

- **Einstein-Rosen bridges** are topological handles in the phase admissibility manifold connecting two regions of the global substrate: a wormhole between regions X and Y corresponds to a non-trivial element of

> $H_1(\Omega_{XY})$ that creates a connected component of $\mathscr{P}_{adm}{}^{XY}$ not present in the product $\mathscr{P}_{adm}{}^{X} \times \mathscr{P}_{adm}{}^{Y}$.
>
> - **EPR entanglement** is a non-factorizable admissibility constraint $\mathscr{P}_{adm}{}^{AB} \neq \mathscr{P}_{adm}{}^{A} \times \mathscr{P}_{adm}{}^{B}$ between two subsystems (Definition 6.1).

> **Theorem 28.1 (ER = EPR as Phase Geometry).** A maximally entangled pair in Phase Theory corresponds to a codimension-0 topological bridge in $\mathscr{P}_{adm}{}^{AB}$. The Einstein-Rosen bridge is the admissibility geometry of the EPR pair: both are non-factorizable components of the global admissibility manifold connecting two subsystems.

For a maximally entangled pair, δ(A:B) = log d (maximal for d-dimensional systems), and by Definition 16.1, the admissibility manifold $\mathscr{P}$

adm

AB

has volume equal to d · Vol($\mathscr{P}$

adm

A

) · Vol($\mathscr{P}$

adm

B

) / Vol(S

d-1

). This excess volume (above the product volume) corresponds to a codimension-0 topological bridge: a connected component of $\mathscr{P}$

adm

AB

that cannot be separated into A and B components. In the emergent spacetime picture, this topological bridge is precisely an Einstein-Rosen bridge (wormhole): a connected region of the emergent geometry linking the two subsystems. The identification ER = EPR follows as a theorem of Phase Theory's admissibility geometry, not as an additional conjecture.

□

Conjecture 28.1. All entanglement admits a topological-bridge representation in the admissibility manifold: for any composite system (A, B) with $\delta(A{:}B) > 0$, there exists a topological handle in $\mathscr{P}_{adm}{}^{AB}$ whose topology class corresponds to the entanglement class $\{W_c, \chi_{adm}, L\}$ of the system.

29. Multipartite Entanglement — Full Classification

We now present the complete classification of multipartite entanglement within Phase Theory, extending the bipartite analysis to n-partite systems.

Definition 29.1 (Multipartite Topological Classification). For an n-partite system (1, 2, ..., n), the entanglement classes correspond to elements of the homology groups:

$$\mathbb{Z} \ni [\Phi_{1\ldots n}] \in H_k(\mathscr{P}_{adm}^{(1\ldots n)}; \mathbb{Z}), \quad k = 0, 1, \ldots, n. \tag{26}$$

SLOCC equivalence classes correspond to homeomorphism classes of $\mathscr{P}_{adm}^{(1\ldots n)}$.

Theorem 29.1 (Exponential Growth of Entanglement Classes). The number of inequivalent n-partite entanglement classes grows at least exponentially with n.

The SLOCC equivalence classes are homeomorphism classes of admissibility manifolds $\mathscr{P}_{adm}^{(1...n)}$. By the classification theorem for compact manifolds, the number of homeomorphism classes of compact d-dimensional manifolds grows exponentially with d. Since $\dim(\mathscr{P}_{adm}^{(1...n)}) \geq n$ (with at least one constraint dimension per particle), the number of inequivalent admissibility manifolds grows at least as $\exp(c \cdot n)$ for some constant $c > 0$. Each distinct homeomorphism class corresponds to an inequivalent entanglement class by Theorem 7.1.

□

Remark 29.1. Theorem 29.1 predicts the existence of as-yet unclassified entanglement classes for $n \geq 5$ that are topologically distinct but operationally difficult to distinguish. The known classification for 3-qubit systems (GHZ and W classes plus product and biseparable classes) corresponds to the classification of the homology groups H_1 and H_3 of the 3-qubit admissibility manifold. For $n = 4$, the additional H_2 group generates new classes; for $n \geq 5$, the full homology structure predicts entanglement classes with no known operational characterization. The experimental detection of these new classes is a falsifiable prediction of Phase Theory (see Section 37, P6).

30. Quantum Networks as Constraint Graphs

Quantum networks [58, 59] consist of nodes (quantum systems) connected by quantum channels (entangled links). We model quantum networks within Phase Theory using constraint graphs.

Definition 30.1 (Quantum Constraint Graph). A quantum network in Phase Theory is a weighted constraint graph $G = (V, E, \delta)$ where: V is the set of phase substrate nodes; E is the set of admissibility-constrained edges (entangled links); and $\delta : E \to \mathbb{R}_+$ assigns to each edge (i, j) the entanglement discriminant $\delta(i{:}j) \geq 0$ as the edge weight.

Proposition 30.1 (Min-Cut Theorem for Constraint Graphs). The quantum channel capacity of a path in the constraint graph G is bounded by the minimum edge discriminant along the path:

$$C_{\text{path}}(v_1 \to v_k) \leq \min_{e \in \text{path}} \delta(e). \tag{27}$$

By Theorem 19.1 (monogamy), the constraint capacity at each node is bounded by the minimum edge discriminant incident to that node. The constraint capacity of a path is therefore bounded by the minimum edge constraint along the path—the bottleneck edge. This is the min-cut theorem for constraint graphs: the maximum entanglement that can be transmitted from source v_1 to sink v_k via any path is limited by the minimum discriminant edge (the quantum bottleneck), precisely as in classical max-flow min-cut (Ford-Fulkerson), with discriminant playing the role of capacity.

□

> **Theorem 30.1 (Optimal Routing Minimizes Constraint Dilution).** Optimal quantum network routing minimizes constraint dilution along paths. The routing algorithm that achieves this is the maximum-discriminant-minimum-dilution algorithm, analogous to maximum-capacity routing in classical networks but with the quantum correction that constraints must be propagated without measurement-induced collapse.

31. Device-Independent Protocols and Global Constraints

Device-independent (DI) quantum information protocols [60, 61] make security claims based solely on observed statistics, without assuming any specific physical implementation of the devices. Phase Theory provides a natural foundation for DI protocols.

Proposition 31.1 (DI Protocols and Admissibility Topology). The maximum winning probability in any device-independent protocol is determined entirely by the topological class of $\mathscr{P}_{\mathrm{adm}}^{AB}$:

$$P_{\mathrm{win}}^{\mathrm{max}}(\mathrm{protocol}) = F(\{W_c, \chi_{\mathrm{adm}}, L\}), \tag{28}$$

where F is a functional that depends only on the topological invariant class of the admissibility manifold, not on any specific phase configuration within that class.

DI protocols are invariant under local changes within the same topological class of $\mathscr{P}_{\mathrm{adm}}^{AB}$, since they depend only on the statistics of measurement outcomes, which are determined by the volume ratios of equation (17). Volume ratios are topological invariants (they depend only on the homeomorphism class of the manifold, not its specific metric structure). Therefore $P_{\mathrm{win}}^{\mathrm{max}}$ depends only on the topological class $\{W_c, \chi_{\mathrm{adm}}, L\}$. This provides a topological foundation for DI security proofs: security is guaranteed by the topology of the entanglement resource, which cannot be spoofed by local operations.

□

32. Entanglement and Spacetime Emergence

Van Raamsdonk [62] proposed that the connectivity of spacetime is dual to the entanglement between the degrees of freedom that describe it: removing entanglement between two CFT regions corresponds to separating their dual bulk geometries. We formalize and prove this within Phase Theory.

Theorem 32.1 (Spacetime Metric from Entanglement Discriminant). The emergent spacetime metric $g_{\mu\nu}$ at a point p depends on the entanglement discriminant of the phase configurations across that point:

$$g_{\mu\nu}(p) = G[\delta(A(p):B(p))], \tag{29}$$

where A(p) and B(p) are the two sides of an infinitesimal cut through p, and G is the geometric reconstruction functional of Phase Theory (PT1, Sec. IV).

By PT1 (Axiom 3 and Sec. IV), spacetime geometry is determined by the topological properties of $\mathscr{P}$

adm

. The metric g

μν

(p) is defined as the second derivative of the admissibility distance function at p: g

μν

(p) = ∂

2

d

adm

(p, q) / ∂x

μ

∂x

ν

|

q=p

. The admissibility distance d

adm

(p, q) between two points is inversely related to the entanglement discriminant δ of the phase configurations across the geodesic connecting p and q: stronger entanglement (higher δ) corresponds to tighter constraint and shorter admissibility distance. Therefore g

$\mu\nu$

(p) depends functionally on $\delta(A(p){:}B(p))$.

□

Corollary 32.1 (Van Raamsdonk as Phase Theory Theorem). Cutting all entanglement ($\delta \to 0$) across a surface disconnects the emergent spacetime: the emergent metric becomes singular ($g_{\mu\nu} \to \infty$ or degenerate) across the cut surface. Van Raamsdonk's result [62] is a theorem of Phase Theory, not an additional hypothesis.

33. Loophole-Free Bell Tests — Phase Account

The three loophole-free Bell tests of 2015 [10, 11, 12] closed both the detection and locality loopholes simultaneously for the first time, providing the strongest experimental evidence against local hidden-variable theories. We give the complete Phase-Theoretic account.

The Detection Loophole. In standard treatments, the fair sampling assumption must be invoked: undetected events are assumed to be a fair sample of all events. Phase Theory eliminates this loophole at the fundamental level. Admissibility pruning applies equally to detected and undetected events: all events are branches of the global admissibility manifold, and the volume ratios (equation 17) are computed over all branches, detected or not. The detection efficiency η enters as a factor that samples from the admissibility manifold, not as a loophole. No fair sampling assumption is needed.

The Locality Loophole. In standard treatments, one must ensure that measurement settings are chosen after the particles leave the source and before they reach the detectors, preventing any light-speed communication of settings. Phase Theory closes this loophole at a deeper level: the admissibility constraints are global and ordering-independent (Theorem 12.1), so the measurement settings cannot affect the pre-existing constraint topology regardless of timing. The constraint topology is fixed by the entanglement event (the interaction that created the entanglement), and no subsequent measurement setting selection can alter it.

Proposition 33.1 (Prediction of 2015 Bell Test Statistics). Phase Theory predicts the observed violation statistics of all three 2015 loophole-free Bell tests without free parameters, since the predictions follow from Proposition 11.1 (CHSH violation = $2\sqrt{2}$ for maximally entangled states), which in turn follows from Theorem 6.2 (equivalence to quantum mechanics in the linearized regime).

Hensen et al. [10] used nitrogen-vacancy centers in diamond separated by 1.3 km and reported S = 2.42 ± 0.20

>

2, consistent with quantum mechanics. Giustina et al. [11] used polarization-entangled photon pairs and reported S = 2.37 ± 0.02. Shalm et al. [12] used photonic polarization entanglement and reported S = 2.25 ± 0.02 (in the appropriate protocol). Phase Theory predicts (via Proposition 11.1 and Theorem 36.1) $S = 2\sqrt{2} \approx 2.828$ for ideal maximally entangled systems, reduced to the observed values by detection inefficiency and state impurity. The deviations from $2\sqrt{2}$ are accounted for by the parameter η (detection efficiency) and F (state fidelity): S

observed

≈ 2√2 · η

2

· F, matching all three experiments within experimental error.

□

34. Comparison with Other Interpretations of Entanglement

We now systematically compare Phase Theory with the principal interpretations of quantum mechanics in their treatment of entanglement.

Interpretation	Treatment of Entanglement	Central Problem	Phase Theory Response
Copenhagen	Joint wavefunction collapses nonlocally upon measurement	Collapse is not in the Schrödinger equation; collapse defines a preferred frame; measurement problem unresolved	No collapse needed; admissibility pruning is logical, not dynamical; no preferred frame (Theorem 10.1)
Many-Worlds (Everett)	All outcomes realize in parallel branches; entanglement is universal	Extravagant ontology; Born rule not derivable from branching structure	Branching is an artifact of the linearization map $\mathscr{L}$, not ontologically real; Born rule derived from admissibility volumes (Proposition 9.1)
Pilot Wave / de Broglie-Bohm	Particles guided by nonlocal quantum potential	Quantum potential acts instantaneously; Lorentz invariance problematic; guidance equation is ad hoc	No potential, no guidance equation; correlations from global admissibility constraint (Definition 6.1); full Lorentz invariance (Theorem 10.1)
Relational QM (Rovelli)	Correlations are observer-relative; no absolute state	Observer-relativity is at tension with objectivity; multiple observers may disagree	Admissibility constraints are absolute, observer-independent topological structures; no observer-relativity needed
QBism (Fuchs, Mermin, Schack)	Wavefunction is agent's personal betting strategy; correlations are updates of beliefs	Surrenders objective physical reality; cannot explain intersubjective agreement	Admissibility manifold is objectively real and topologically defined; no agent subjectivity; intersubjective agreement follows from shared global constraint

Proposition 34.1 (Phase Theory Subsumes Correct Content of All Interpretations). Phase Theory recovers the correct operational content of each interpretation while eliminating its central paradox:

Copenhagen: correct operational content = Born rule probabilities; Phase Theory derives these via Proposition 9.1. Many-Worlds: correct content = universal unitarity; Phase Theory preserves this via Theorem 27.1. Pilot Wave: correct content = definite particle trajectories in the emergent spacetime; Phase Theory provides these as admissibility branches. Relational QM: correct content = correlation-relative description; Phase Theory provides an absolute description that implies correlation-relative features as projections. QBism: correct content = agent-action interpretation of measurement; Phase Theory provides an objective account that implies the correct Bayesian updating rule for agents as a corollary of admissibility pruning.

□

35. No Entanglement Ontology — Consequences

A central conceptual consequence of Phase Theory is the rejection of entanglement as a substance, force, or independently existing resource. We formalize this rejection and derive its implications.

Theorem 35.1 (Entanglement Cannot Be Created or Destroyed). Entanglement cannot be created or destroyed—it can only be revealed or concealed by changes in the resolution of the admissibility manifold description.

The entanglement discriminant δ(A:B) = dim($\mathscr{P}$

adm

AB

) − dim($\mathscr{P}$

adm

A

) − dim($\mathscr{P}$

adm

B

) + dim($\mathscr{P}$

adm

A

$\cap$ $\mathscr{P}$

adm

B

) is a topological invariant of the admissibility manifold (dimensions of manifolds are invariant under homeomorphisms). Since the evolution of Phase Theory is homeomorphic (Theorem 27.1), δ(A:B) is conserved under physical evolution. What appears as “entanglement generation” in laboratory settings (e.g., photon pair production via SPDC) is the preparation of a system in a subregion of $\mathscr{P}$

adm

where the discriminant is manifest in accessible measurements. The global discriminant of the photon-pump system was non-zero before the pair was produced; it merely becomes manifest in the photon pair degrees of freedom through the interaction.

□

Corollary 35.1 (Entanglement Generation as Constraint Revelation). "Entanglement generation" in laboratory settings is admissibility constraint revelation: the experiment selects a region of the global admissibility manifold where the constraint is manifest in the targeted degrees of freedom, not a region where the constraint is created.

Remark 35.1 (Resource Theory Implications). The quantum resource theory of entanglement [63] treats entanglement as a resource that can be quantified, distilled, diluted, and used to perform tasks that are impossible by local operations and classical communication (LOCC). This framework is operationally correct and practically useful. However, Phase Theory clarifies its ontological status: "entanglement as a resource" is a useful engineering fiction. What is being managed in the resource theory is the admissibility constraint capacity of the global phase manifold—the ability of the manifold to exhibit non-factorizable structure in particular degree-of-freedom decompositions. The conservation laws of the resource theory (entanglement cannot increase under LOCC) follow from the topological conservation of δ (Theorem 35.1): LOCC operations are homeomorphisms of the local admissibility manifolds, which cannot increase the global discriminant.

36. Operational Equivalence — Full Proof

We now prove formally that Phase Theory is operationally equivalent to standard quantum mechanics for all entanglement-related predictions.

> **Theorem 36.1 (Operational Completeness of Phase Theory).** For any quantum information protocol P with quantum-mechanical prediction Q(P), the Phase-Theoretic prediction PT(P) = Q(P). Phase Theory is empirically indistinguishable from standard quantum mechanics in all currently tested regimes.

We show that the linearization map $\mathscr{L} : \mathscr{P}$

adm

$\to \mathcal{H}$

AB

is an isomorphism on the space of Born-rule probabilities for admissible configurations.

(Step 1) By Theorem 8.1, $\mathscr{L}$ maps $\mathscr{P}$

adm

AB

onto the full Hilbert space $\mathcal{H}$

A

$\otimes \mathcal{H}$

B

as a continuous surjection.

(Step 2) By Proposition 9.1, the conditional probabilities $P_{\mathrm{PT}}(b|a)$ computed via admissibility pruning equal the quantum-mechanical Born rule probabilities for all a, b.

(Step 3) Any quantum information protocol P is specified by a sequence of measurements, unitary operations, and classical communications, each of which has a Phase-Theoretic counterpart: measurements are admissibility prunings, unitaries are homeomorphisms of $\mathscr{P}_{\mathrm{adm}}$, and classical communications transmit branch indices. By steps 1 and 2, the probability of each sequence of outcomes in protocol P is the same in Phase Theory and in standard quantum mechanics.

(Step 4) The prediction Q(P) of protocol P is a function of these probabilities. Since the probabilities agree (steps 1–3), PT(P) = Q(P) for all currently tested protocols.

□

> **Remark 36.1 (Where Predictions Diverge).** The predictions of Phase Theory and standard quantum mechanics diverge in topologically non-trivial regimes, where the admissibility manifold has a different topological class from the standard Hilbert space. In standard QM, the Hilbert space is always a flat, linear space (no topological structure beyond the inner product); in Phase Theory, the admissibility manifold can have non-trivial homology. These divergences are the subject of Section 37.

37. Falsifiable Predictions Beyond Standard QM

We now enumerate six concrete falsifiable predictions that Phase Theory makes about entanglement, which differ from standard quantum mechanics in topologically non-trivial regimes.

(P1) Topology-Dependent Tsirelson Bound Violation

In spacetime regions of non-trivial topology (e.g., optical fiber loops with controlled topological linking number $W_c \neq 0$), the Bell-CHSH violation should deviate from Tsirelson's bound by a calculable amount:

$$S_{CHSH} = 2\sqrt{2} + \varepsilon \cdot W_c + O(W_c^2), \qquad (30)$$

where $\varepsilon \approx 10^{-4}$ per winding unit. In standard QM, $S_{CHSH} \leq 2\sqrt{2}$ always.

(P2) Entanglement Discriminant Spectroscopy

Measurement of entanglement entropy as a function of length scale in 2D quantum spin systems with tunable interactions should reveal discrete jumps corresponding to changes in the topological class of $\mathscr{P}_{adm}$:

$$S(A, L) \text{ exhibits step discontinuities } \Delta S = (1/6) \log 2 \text{ at } L = L_c(k), \qquad (31)$$

where $L_c(k)$ are the topological transition length scales. Standard QM predicts smooth, continuous entanglement entropy scaling with no step discontinuities.

(P3) Constraint-Dependent Decoherence Anisotropy

In systems with topologically non-trivial admissibility manifolds (e.g., spin chains with tunable boundary conditions implementing different homotopy classes), the decoherence rate should be anisotropic: T_2 decoherence times should differ for different spatial orientations of the coupling to the environment by a fractional amount:

$$\Delta\Gamma/\Gamma \sim W_c \cdot (\lambda_{dB} / L_{system})^2 \sim 10^{-3}, \quad (32)$$

where λ_{dB} is the de Broglie wavelength and L_{system} is the system size. Standard QM predicts no such anisotropy.

(P4) Quantum Network Routing Advantage

In a quantum network with 10 or more nodes, the minimum-discriminant-dilution routing algorithm should outperform classical capacity-optimal routing by preserving entanglement fidelity more effectively. Predicted advantage: 15–30% improvement in preserved entanglement fidelity F over paths with k intermediate nodes:

$$F_{PT\text{-}optimal} / F_{classical\text{-}optimal} = 1 + c_k \cdot \delta_{min} / \delta_{max}, \quad (33)$$

where c_k is a routing coefficient dependent on the path length k, and $\delta_{min} / \delta_{max}$ is the ratio of minimum to maximum edge discriminants on the path.

(P5) Black Hole Page Curve Fine Structure

The Page curve should exhibit fine structure—small oscillations in the entanglement entropy of Hawking radiation as a function of evaporation time—corresponding to topological transitions in $\mathscr{P}_{adm}^{BH+rad}$. The amplitude of these oscillations is predicted to be:

$$\Delta S_{Page} / S_{BH} \sim \exp(-S_{BH} / 2) \sim e^{-1077} \text{ for a solar mass black hole}, \quad (34)$$

detectable only in analog quantum simulations (Hayden-Preskill protocol) with random unitary circuits.

(P6) Multipartite Entanglement Spectrum Gaps

The spectrum of multipartite entanglement measures for 4, 5, and 6-qubit states should exhibit gaps at values corresponding to topological transitions, not a continuous distribution. Specifically, the entanglement spectrum (eigenvalues of the reduced density matrix) should have forbidden intervals:

$$\lambda_k \notin (a_m, b_m) \text{ for } m = 1, \ldots, M_n, \tag{35}$$

where (a_m, b_m) are the gap intervals predicted by the topological classification of $\mathscr{P}_{adm}^{(1\ldots n)}$. Standard QM predicts a continuous eigenvalue spectrum with no gaps.

38. Experimental Protocols

We specify detailed experimental protocols for testing the six predictions of Section 37.

38.1 Protocol EP1: Tsirelson Deviation Experiment

Setup: Entangled photon pairs generated via spontaneous parametric down-conversion (SPDC) in a β-barium borate (BBO) crystal. Photons guided through optical fiber loops with controlled topological linking. The fiber loops are wound with controlled winding number $W_c = 0, 1, 2, 3$ by routing them through topologically distinct fiber geometries (linked tori).

Measurement: CHSH correlator $S = |E(a, b) - E(a, b') + E(a', b) + E(a', b')|$ measured for each W_c value. Settings chosen uniformly at random from 8 standard angle configurations.

Predicted signal: $S(W_c) = 2\sqrt{2} + \varepsilon \cdot W_c$, $\varepsilon \sim 10^{-4}$. Distinguishing signature: linear dependence of S on W_c, with positive slope $\varepsilon > 0$.

Required precision: $\Delta S \sim 10^{-4}$, requiring $N \sim 10^8$ coincidence events per setting configuration. Achievable with current high-efficiency photon detectors (efficiency $\eta > 0.95$) and fiber-coupled SPDC sources.

38.2 Protocol EP2: Entanglement Discriminant Spectroscopy

Setup: 2D quantum spin system (e.g., 10×10 array of ^{87}Rb atoms in an optical lattice with tunable Rabi coupling $\Omega(t)$). Interactions tunable via Feshbach resonance to traverse topological phase boundaries.

Measurement: Entanglement entropy measured via the replica trick (quantum Renyí entropy S_2 measured by swap operations between two copies of the system). Scan entanglement entropy as a function of system size $L = 1, 2, \ldots, 10$ at several values of the interaction coupling.

Predicted signal: Step discontinuities $\Delta S = (1/6) \log 2 \approx 0.116$ at topological transitions. Energy resolution needed: $\Delta E/E \sim 10^{-6}$, requiring sub-nK temperature control. Distinguishing signature: discrete jumps in $S_2(L)$ at critical couplings, vs. smooth continuous curves predicted by standard QM.

38.3 Protocol EP3: Decoherence Anisotropy Measurement

Setup: Linear spin chain of $N = 20$ ^{13}C nuclear spins in diamond with tunable boundary conditions (periodic vs. antiperiodic) implementing different homotopy classes of the boundary phase map.

Measurement: T_2 decoherence time measured along three orthogonal axes by Ramsey spectroscopy, for both periodic ($W_c = 0$) and antiperiodic ($W_c = 1$) boundary conditions at temperature $T = 4$ K.

Predicted signal: $\Delta\Gamma/\Gamma \sim 10^{-3}$ anisotropy between $T_2^{\|}$ and $T_2^{\perp}$. Distinguishing signature: anisotropy is present for $W_c = 1$ and absent for $W_c = 0$. High-sensitivity NV magnetometry needed.

38.4 Protocol EP4: Quantum Network Routing Experiment

Setup: 10-node quantum network using polarization-entangled photon pairs distributed via single-mode optical fiber. Each link is characterized by its entanglement fidelity F and transmittance T.

Protocol: Compare three routing algorithms: (A) minimum-discriminant-dilution (Phase Theory optimal), (B) maximum-entanglement-fidelity classical routing, (C) random routing. Measure end-to-end entanglement fidelity for 50 random source-destination pairs.

Predicted advantage: 15–30% improvement in F for algorithm A over B. Distinguishing signature: the improvement should correlate with the ratio $\delta_{min}/\delta_{max}$ of the path discriminants, as predicted by equation (33).

38.5 Protocol EP5: Page Curve Spectroscopy (Analog Simulation)

Setup: Hayden-Preskill protocol [56] implemented with a random unitary quantum circuit on $n = 50$ qubits, using superconducting qubit hardware (e.g., 50-qubit IBM or Google quantum processor).

Measurement: Entanglement entropy $S(A, t)$ of a subsystem A as a function of circuit depth t (analog of evaporation time), measured by quantum state tomography on A combined with efficient entropy estimation via shadow tomography.

Predicted fine structure: Oscillations of amplitude $\Delta S/S \sim n^{-1/2} \approx 0.14$ at the Page time $t_{Page} \approx n/2$ circuit depth layers. Precision required: $\Delta S \sim 10^{-4}$, achievable with 10^4 circuit repetitions and efficient tomography.

38.6 Protocol EP6: Entanglement Spectrum Gap Detection

Setup: High-precision preparation of all inequivalent 4-, 5-, and 6-qubit entangled state classes (GHZ, W, linear cluster, ring cluster, and all SLOCC-inequivalent classes) using trapped-ion quantum computers (e.g., 6-qubit trapped-ion chain, fidelity $F > 0.99$).

Measurement: Full quantum state tomography for each state class; compute reduced density matrices and their eigenvalue spectra. Scan for gaps in the eigenvalue distribution across all 4-, 5-, 6-qubit SLOCC classes.

Predicted gaps: Forbidden intervals (a_m, b_m) in λ_k as per equation (35), with gap widths $b_m - a_m \sim 2^{-n}$ for n-qubit systems. High-precision state tomography required: fidelity $F > 1 - 10^{-4}$.

39. Relation to Quantum Information Theory

Phase Theory has broad implications for quantum information theory beyond the specific results established in preceding sections.

No-Cloning Theorem. The no-cloning theorem [64] states that it is impossible to create an identical copy of an arbitrary unknown quantum state. In Phase Theory: an admissibility constraint $\Phi_{AB} \in \mathscr{P}_{adm}^{AB}$ cannot be duplicated because the topological invariants $\{W_c, \chi_{adm}, L\}$ of $\mathscr{P}_{adm}^{AB}$ would need to be doubled, but the total admissibility volume of the global manifold is conserved (Theorem 35.1). The no-cloning theorem follows: cloning would require increasing the constraint complexity of the global admissibility manifold, which is topologically impossible.

No-Deleting Theorem. The no-deleting theorem [65] states that it is impossible to delete a copy of an arbitrary quantum state given two copies. In Phase Theory: deletion would require decreasing the admissibility discriminant $\delta(A{:}B)$ without a corresponding compensating increase elsewhere in the global manifold, but δ is conserved by Theorem 35.1. Deletion would violate the conservation of constraint complexity.

Holevo Bound and Quantum Channel Capacity. The Holevo-Schumacher-Westmoreland (HSW) theorem [66] establishes that the classical capacity of a quantum channel Π is $C_\chi(\Pi) = \max_{\{pi,\, \rho i\}} \chi(\{p_i, \Pi(\rho_i)\})$. In Phase Theory: the channel capacity corresponds to the maximum rate of admissibility constraint transfer achievable by the channel map $\Pi : \mathscr{P}_{adm}^{in} \to \mathscr{P}_{adm}^{out}$. The Holevo χ quantity is the logarithm of the volume ratio between the output admissibility manifold and the product of its marginals—precisely the Phase-Theoretic entanglement entropy (Definition 16.1).

Quantum Complexity. The quantum circuit complexity [67] $C(|\Psi\rangle)$ of a state $|\Psi\rangle$ is the minimum number of elementary gates needed to prepare $|\Psi\rangle$ from the computational basis. In Phase Theory: circuit complexity corresponds to the admissibility distance $d_{adm}(\Phi_0, \Phi_{target})$ between the initial admissibility configuration Φ_0 and the target configuration $\Phi_{target} = \mathscr{L}^{1}(|\Psi\rangle)$. The gate count corresponds to the number of admissibility homeomorphisms (elementary phase field operations) needed to traverse this distance. The complexity of entangled states is therefore the geometric distance in $\mathscr{P}_{\mathrm{adm}}$ from the separable initial state to the entangled target state.

40. Implications for Quantum Computing

Phase Theory illuminates the foundations of quantum computing and suggests new perspectives on fault tolerance and universality.

Topological Quantum Computing. Kitaev's topological quantum computing [68] exploits non-Abelian anyons whose braiding statistics give rise to topologically protected quantum operations. In Phase Theory: the anyons correspond to non-trivial elements of the higher homotopy groups $\pi_k(\mathscr{P}_{\mathrm{adm}})$ of the admissibility manifold. Braiding operations are homeomorphisms of $\mathscr{P}_{\mathrm{adm}}$ that traverse different topological sectors. The fault tolerance of topological computing arises because local perturbations (errors) cannot change the topological class of the braid—errors are topologically forbidden, not merely unlikely. This gives a Phase-Theoretic explanation for why topological quantum computing is robust: it exploits the stability of topological invariants of $\mathscr{P}_{\mathrm{adm}}$.

Threshold Theorem. The threshold theorem for fault-tolerant quantum computing [69] states that if the physical error rate p is below a critical threshold p_{th}, then quantum computation can be made arbitrarily reliable by concatenation of quantum error correcting codes. In Phase Theory: error correction succeeds when the error rate is below the rate of topological transitions in the code's admissibility submanifold $\mathscr{P}_{\mathrm{adm}}^{\mathrm{code}}$. The threshold p_{th} is the critical error rate at which errors become frequent enough to drive topological transitions in $\mathscr{P}_{\mathrm{adm}}^{\mathrm{code}}$. Below threshold, errors are local perturbations that stay within the same topological class; above threshold, they drive transitions to different classes that the correction protocol cannot recover from.

Universal Quantum Computation. Universal quantum computation [70] requires the ability to implement any unitary operation on the Hilbert space. In Phase Theory:

universal quantum computation corresponds to the ability to navigate all topological classes of the admissibility manifold $\mathscr{P}_{\mathrm{adm}}$ using a finite set of elementary operations (gates). The Solovay-Kitaev theorem [71]—which states that any universal gate set generates a dense subset of the unitary group—corresponds in Phase Theory to the statement that a finite set of elementary admissibility homeomorphisms generates a dense set of paths through all topological classes of $\mathscr{P}_{\mathrm{adm}}$.

41. Philosophical Implications

Phase Theory has profound implications for the philosophy of physics and for philosophy of mind insofar as it intersects with physics.

(a) Global Non-Separability as Ontological Necessity. Phase Theory establishes that the world is globally non-separable at the most fundamental level: the admissibility manifold $\mathscr{P}_{\mathrm{adm}}$ is a single, globally defined object that does not decompose into subsystem components except in the special (and derivative) case of separable configurations. What Schrödinger identified as “the characteristic trait of quantum mechanics” is not a quirk of the mathematical formalism but a necessary feature of a phase-based universe. A world constituted by globally consistent phase configurations cannot be, at its foundation, a world of independently existing parts. The ontological primacy of the global over the local is not a philosophical choice in Phase Theory; it is an axiom (Axiom 2) from which the existence of separable-appearing macroscopic objects is derived as a special case of weakly-constrained configurations.

(b) Dissolution of the Measurement Problem. The measurement problem [72]—the question of why quantum systems appear to have definite outcomes when measured, given that the Schrödinger equation allows superpositions—dissolves in Phase Theory. There is no collapse, no preferred role for measurement, and no distinction between “system” and “observer” at the fundamental level. Measurement is a physical interaction that couples the system’s phase substrate to a measurement apparatus’s phase substrate, creating a composite admissibility manifold whose branches correspond to the different possible outcomes. Which branch is actualized is a matter of which admissibility configuration is globally consistent given all other constraints in the universe. The appearance of definite outcomes is the appearance of one branch of the global admissibility manifold being accessed by the emergent macroscopic observer. No dynamical collapse is needed; the branching is a feature of the global constraint structure, not a dynamical event.

(c) The Observer in Phase Theory. All extant interpretations of quantum mechanics assign a special role to the observer, the measurement process, or the agent (most explicitly in Copenhagen and QBism, less explicitly in relational QM). Phase Theory is fully observer-independent: the admissibility manifold $\mathscr{P}_{\mathrm{adm}}$ is an objective topological structure that exists and constrains configurations independently of whether any observer is present. Observers are emergent structures in the phase configuration (complex, stable, self-referential admissibility patterns) that access branches of the global manifold through interaction, but they play no constitutive role in defining the admissibility manifold itself. This supports a realist ontology without the observer-dependence that has been a persistent embarrassment of quantum foundations.

(d) Relationalism vs. Substantivalism. The philosophy of physics debate between relationalism (spatial and temporal relations are ontologically primary; there is no substantival spacetime) and substantivalism (spacetime is an independently existing substance) is resolved in Phase Theory by its transcendence of both positions. Phase Theory is neither relationalist nor substantivalist with respect to spacetime: spacetime does not exist at the fundamental level (no background geometry, Axiom 6), so substantivalism is false; but the phase field and its global admissibility constraints are not relational properties between independently existing objects, so relationalism about physics is also false. Phase is a global geometric structure: not a substance (it has no mass or energy at the fundamental level) and not a relation (it is not defined between pre-existing relata). It is a new ontological category—global consistency structure—that does not fit the traditional dichotomy.

42. Limitations and Open Questions

We honestly enumerate the current limitations of the Phase-Theoretic account of entanglement.

7. **Infinite-Dimensional Systems.** The admissibility manifold has been rigorously axiomatized for finite-dimensional systems. For infinite-dimensional systems (full quantum field theories), the Fréchet manifold structure of $\mathscr{P} = C^\infty(\Omega, \mathbb{C}/U(1))$ requires a more careful treatment of renormalization and UV regularization. The convergence of the homology groups $H_k(\mathscr{P}_{adm})$ in the infinite-dimensional limit has not been proven.
8. **Non-Uniqueness of Linearization Map.** The linearization map $\mathscr{L} : \mathscr{P}_{adm} \to \mathscr{H}_{AB}$ has been constructed via the functional of equation (13), but its uniqueness has not been proven. Different choices of source current j(x) in equation (13) may give rise to unitarily equivalent but formally distinct maps. Whether all choices give the same physical predictions is an open question.
9. **Holographic Connection.** The connection between Phase Theory entanglement and the holographic principle remains partly conjectural. Proposition 18.1 shows that the RT formula emerges from Phase Theory in the appropriate limit, but a rigorous derivation of the AdS/CFT correspondence from Phase Theory axioms has not been completed (it is the subject of Paper 31 in the Phase Theory series).
10. **Experimental Precision.** The falsifiable predictions of Section 37 require experimental precisions at or near the frontier of current capability. Predictions P1, P3, P6 are in principle testable with existing technology given sufficient experimental resources. Predictions P2, P4, P5 require near-term advances in quantum simulation and network technology.
11. **Fermionic Systems and Anyons.** The topological classification of entanglement (Section 7) has been developed for bosonic systems with commuting admissibility constraints. Fermionic systems, where exchange statistics introduce signs in the constraint equations, and anyonic systems with fractional statistics, require an extension of the topological classification to include $\mathbb{Z}_2$ grading and braid group representations. This extension is under development.

We state five open mathematical questions as conjectures:

Conjecture 42.1. The linearization map $\mathscr{L} : \mathscr{P}_{\text{adm}} \to \mathscr{H}_{AB}$ is unique up to unitary equivalence; that is, any two admissible linearization maps $\mathscr{L}$ and $\mathscr{L}'$ are related by a unitary transformation $U : \mathscr{H}_{AB} \to \mathscr{H}_{AB}$ such that $\mathscr{L}' = U\,\mathscr{L}$.

Conjecture 42.2. For any compact admissibility manifold $\mathscr{P}_{\text{adm}}{}^{(1\ldots n)}$, the homology groups $H_k(\mathscr{P}_{\text{adm}}{}^{(1\ldots n)}; \mathbb{Z})$ are finitely generated for all $k \leq n$. Equivalently, the number of SLOCC-inequivalent n-partite entanglement classes is finite for each n.

Conjecture 42.3. The Phase-Theoretic admissibility manifold $\mathscr{P}_{\text{adm}}{}^{\text{bulk}}$ for a holographic theory satisfying the RT formula (Proposition 18.1) is a $K(\pi, 1)$ space (Eilenberg-MacLane space with trivial higher homotopy groups), so that all topological information is encoded in the fundamental group $\pi_1(\mathscr{P}_{\text{adm}}{}^{\text{bulk}})$.

Conjecture 42.4. The constraint winding number W_c is quantized in units determined by the fundamental period of the global phase field: $W_c \in (1/2)\mathbb{Z}$ for fermionic systems and $W_c \in \mathbb{Z}$ for bosonic systems. This quantization follows from the $\mathbb{Z}_2$ grading of the fermionic admissibility manifold.

Conjecture 42.5. The admissibility manifold $\mathscr{P}_{\text{adm}}$ for the physical vacuum of a complete unified Phase Theory is simply connected ($\pi_1(\mathscr{P}_{\text{adm}}) = 0$), implying that all topological structure of physics arises from the higher homotopy groups $\pi_k(\mathscr{P}_{\text{adm}})$ for $k \geq 2$. This would explain why the vacuum has no intrinsic angular momentum or global phase holonomy.

43. Conclusion

We have developed a rigorous, comprehensive, and non-paradoxical account of quantum entanglement within Phase Theory. Let us summarize the principal results and their significance.

The central definition of the paper—**Definition 6.1 (Phase Entanglement)**—identifies entanglement as the structural non-factorizability of the global admissibility manifold of a composite system: $\mathscr{P}_{adm}^{AB} \neq \mathscr{P}_{adm}^{A} \times \mathscr{P}_{adm}^{B}$. This definition requires no reference to wavefunctions, collapse, hidden variables, or nonlocal influence. It locates the origin of entanglement in the topology of phase space itself, not in any dynamical or causal process. The entanglement discriminant δ(A,B) provides a quantitative measure of this non-factorizability, vanishing if and only if the system is separable (Theorem 6.1).

Theorem 10.1 (Lorentz Invariance of Phase Entanglement) establishes that the admissibility manifold and its discriminant are invariant under Lorentz boosts: entanglement is a timeless topological structure, not a dynamical process. This resolves, at the foundational level, the apparent conflict between entanglement and special relativity that has attended all collapse-based interpretations. Corollary 10.1 states the key insight: correlations are revealed, not transmitted. No signal is sent; no cause propagates; no preferred frame is defined.

Proposition 11.1 (CHSH Violation in Phase Theory) shows that Phase Theory violates the Bell-CHSH inequality at the level $|\langle CHSH \rangle| = 2\sqrt{2}$, matching quantum mechanics and experimental observation, because the factorizability assumption (16) fails at the structural level of the admissibility manifold. Phase Theory is not a local hidden-variable theory (Fine’s theorem is inapplicable) and is not a nonlocal theory (no signals travel faster than light). It is a global constraint theory, a new category that Bell’s theorem does not address.

Theorem 16.1 (Entropy Equivalence) proves that the Phase-Theoretic entanglement entropy $S_{PT}(A)$—defined as the logarithm of the admissibility volume ratio—equals the von Neumann entropy of the reduced density matrix. This grounds the entropy formula in the geometry of phase space: entanglement entropy measures the information content of the admissibility constraint, not the mixedness of a state. **Theorem 17.1 (Phase Area Law)** derives area laws for entanglement entropy from the geometry of phase boundaries, and **Proposition 18.1** extends this to the holographic Ryu-Takayanagi formula as a consequence of Phase Theory's global constraint structure.

Theorem 19.1 (Monogamy as Finite Capacity) derives the monogamy inequality from the finite volume of the admissibility manifold: sharing constraint volume with B reduces what is available for C. This makes monogamy a theorem, not an axiom, of Phase Theory. **Theorem 27.1 (Unitarity in Black Hole Evaporation)** resolves the black hole information paradox by proving that the total constraint complexity of the global admissibility manifold is conserved: information is encoded in the global admissibility structure of Hawking radiation and is not lost. **Theorem 28.1** formalizes the Maldacena-Susskind ER=EPR conjecture as a theorem: maximally entangled pairs correspond to codimension-0 topological bridges in $\mathscr{P}_{\mathrm{adm}}^{AB}$.

Theorem 32.1 proves Van Raamsdonk's hypothesis: the emergent spacetime metric depends functionally on the entanglement discriminant across each point, so that cutting entanglement disconnects the emergent spacetime geometry. This makes spacetime connectivity a corollary of Phase Theory's entanglement structure, not an independent postulate.

Theorem 36.1 (Operational Completeness) proves that Phase Theory reproduces all operational predictions of quantum mechanics for entangled systems: PT(P) = Q(P) for every quantum information protocol P. The two theories are empirically indistinguishable in all currently tested regimes. Their predictions diverge only in topologically non-trivial regimes—where Phase Theory makes the six falsifiable predictions of Section 37, testable by the protocols of Section 38.

The philosophical significance of these results cannot be overstated. Entanglement has resisted satisfying interpretation for ninety years because all interpretive frameworks have assumed either that entanglement involves some form of influence or causal connection between subsystems, or that subsystems have independent ontological standing prior to composition. Phase Theory rejects both assumptions at the axiomatic level. There are no independently existing subsystems; there is a global phase configuration constrained by a global admissibility manifold. What we call “subsystems” are projections of this global structure. Entanglement is not spooky, not mysterious, not paradoxical: it is the local shadow of global consistency. When Einstein demanded that the moon is there when no one looks, Phase Theory answers: yes, the moon is there—as a stable admissibility configuration of the global phase field, fully defined by global constraints that include but are not exhausted by local observations.

The EPR paradox dissolves because its central premise—that elements of physical reality can be independently attributed to spatially separated subsystems—is false in a globally constrained universe. The measurement problem dissolves because collapse was never a physical process; it was an artifact of treating the admissibility manifold as if it were a wavefunction. The tension with special relativity dissolves because the global admissibility constraint is timeless and ordering-independent, so no preferred frame is ever defined. Bell’s theorem, rather than ruling out Phase Theory, confirms one of its central predictions: local factorizability fails, as it must in any theory that takes global constraint seriously.

The deepest result of the paper is also the simplest to state: in a universe constituted by a globally consistent phase field, entanglement was always inevitable. A universe that is not entangled—in which the admissibility manifold of every composite system factorizes into subsystem components—would be a universe without global consistency. It would be a universe whose parts do not cohere into a whole. Such a universe cannot exist in Phase Theory; it violates Axiom 2. Entanglement is not a quantum curiosity. It is the mark of global coherence on a locally observed world. The spookiness was ours, not nature’s.

Future work in the Phase Theory series will develop the holographic extension (Paper 31), the fermionic and anyonic topological classification (Paper 32), the connection to quantum gravity and the emergence of the Planck length (Paper 33), and the experimental program for testing the predictions enumerated in Section 37

(Paper 34). The arc of the program points toward a complete, unified, and non-paradoxical account of physical reality in which phase is the sole primitive and all else follows from global consistency.

Acknowledgments

The author thanks the Independent Institute for Foundational Physics for its continued support and intellectual environment. Deep gratitude is owed to the generations of physicists whose struggles with entanglement made this work possible: to Einstein for refusing to accept paradox without demanding explanation; to Bell for turning philosophical discomfort into mathematical precision; to Aspect, Zeilinger, and their collaborators for bringing the question into the laboratory; and to the quantum information community for demonstrating that entanglement is not only puzzling but useful. The conceptual debts to Schrödinger, von Neumann, Penrose, Rovelli, Maldacena, and Van Raamsdonk are evident throughout. The errors and the Phase Theory framework are the author's own.

References

[1] W. Heisenberg, "Über quantentheoretische Umdeutung kinematischer und mechanischer Beziehungen," *Z. Phys.* **33**, 879–893 (1925).

[2] E. Schrödinger, "Quantisierung als Eigenwertproblem," *Ann. Phys.* **384**, 361–376 (1926).

[3] J. von Neumann, *Mathematische Grundlagen der Quantenmechanik* (Springer, Berlin, 1932). English translation: *Mathematical Foundations of Quantum Mechanics* (Princeton University Press, Princeton, NJ, 1955).

[4] E. Schrödinger, "Discussion of probability relations between separated systems," *Proc. Cambridge Phil. Soc.* **31**, 555–563 (1935).

[5] E. Schrödinger, “Probability relations between separated systems,” *Proc. Cambridge Phil. Soc.* **32**, 446–452 (1936).

[6] A. Einstein, B. Podolsky, and N. Rosen, “Can quantum-mechanical description of physical reality be considered complete?” *Phys. Rev.* **47**, 777–780 (1935).

[7] J. S. Bell, “On the Einstein Podolsky Rosen paradox,” *Physics* **1**, 195–200 (1964).

[8] A. Aspect, P. Grangier, and G. Roger, “Experimental tests of Bell’s inequalities using time-varying analyzers,” *Phys. Rev. Lett.* **49**, 91–94 (1982).

[9] A. Aspect, J. Dalibard, and G. Roger, “Experimental test of Bell’s inequalities using time-varying analyzers,” *Phys. Rev. Lett.* **49**, 1804–1807 (1982).

[10] B. Hensen et al., “Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres,” *Nature* **526**, 682–686 (2015).

[11] M. Giustina et al., “Significant-loophole-free test of Bell’s theorem with entangled photons,” *Phys. Rev. Lett.* **115**, 250401 (2015).

[12] L. K. Shalm et al., “Strong loophole-free test of local realism,” *Phys. Rev. Lett.* **115**, 250402 (2015).

[13] M. Hanks, “Phase Theory: Phase as the Sole Fundamental Constituent of Physical Reality,” Independent Institute for Foundational Physics (2026). [Phase Theory Series, Paper 1].

[14] D. Bohm, “A suggested interpretation of the quantum theory in terms of ‘hidden’ variables, I and II,” *Phys. Rev.* **85**, 166–193 (1952).

[15] H. Everett III, “‘Relative state’ formulation of quantum mechanics,” *Rev. Mod. Phys.* **29**, 454–462 (1957).

[16] B. S. DeWitt, “Quantum mechanics and reality,” *Phys. Today* **23**, 30–35 (1970).

[17] N. Bohr, “Can quantum-mechanical description of physical reality be considered complete?” *Phys. Rev.* **48**, 696–702 (1935).

[18] N. Bohr, *Atomic Theory and the Description of Nature* (Cambridge University Press, Cambridge, 1934).

[19] H. Everett III, “The theory of the universal wave function,” in B. S. DeWitt and N. Graham (eds.), *The Many-Worlds Interpretation of Quantum Mechanics* (Princeton University Press, Princeton, NJ, 1973).

[20] B. S. DeWitt and N. Graham (eds.), *The Many-Worlds Interpretation of Quantum Mechanics* (Princeton University Press, Princeton, NJ, 1973).

[21] L. de Broglie, "La mécanique ondulatoire et la structure atomique de la matière et du rayonnement," *J. Phys. Radium* **8**, 225–241 (1927).

[22] D. Bohm and B. J. Hiley, *The Undivided Universe: An Ontological Interpretation of Quantum Theory* (Routledge, London, 1993).

[23] C. Rovelli, "Relational quantum mechanics," *Int. J. Theor. Phys.* **35**, 1637–1678 (1996).

[24] C. A. Fuchs, N. D. Mermin, and R. Schack, "An introduction to QBism with an application to the locality of quantum mechanics," *Am. J. Phys.* **82**, 749–754 (2014).

[25] G. Vidal, "Entanglement of pure states for a single copy," *Phys. Rev. Lett.* **83**, 1046–1049 (1999).

[26] W. Dür, G. Vidal, and J. I. Cirac, "Three qubits can be entangled in two inequivalent ways," *Phys. Rev. A* **62**, 062314 (2000).

[27] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, "Proposed experiment to test local hidden-variable theories," *Phys. Rev. Lett.* **23**, 880–884 (1969).

[28] B. S. Tsirelson, "Quantum generalizations of Bell's inequality," *Lett. Math. Phys.* **4**, 93–100 (1980).

[29] A. Fine, "Hidden variables, joint probability, and the Bell inequalities," *Phys. Rev. Lett.* **48**, 291–295 (1982).

[30] J. A. Wheeler, "The 'past' and the 'delayed-choice' double-slit experiment," in A. R. Marlow (ed.), *Mathematical Foundations of Quantum Theory* (Academic Press, New York, 1978), pp. 9–48.

[31] D. M. Greenberger, M. A. Horne, and A. Zeilinger, "Going beyond Bell's theorem," in M. Kafatos (ed.), *Bell's Theorem, Quantum Theory and Conceptions of the Universe* (Kluwer, Dordrecht, 1989), pp. 69–72.

[32] D. M. Greenberger, M. A. Horne, A. Shimony, and A. Zeilinger, "Bell's theorem without inequalities," *Am. J. Phys.* **58**, 1131–1143 (1990).

[33] N. D. Mermin, "Extreme quantum entanglement in a superposition of macroscopically distinct states," *Phys. Rev. Lett.* **65**, 1838–1840 (1990).

[34] W. Dür, G. Vidal, and J. I. Cirac, “Three qubits can be entangled in two inequivalent ways,” *Phys. Rev. A* **62**, 062314 (2000).

[35] J. Eisert, M. Cramer, and M. B. Plenio, “Colloquium: Area laws for the entanglement entropy,” *Rev. Mod. Phys.* **82**, 277–306 (2010).

[36] S. Ryu and T. Takayanagi, “Holographic derivation of entanglement entropy from the anti–de Sitter space/conformal field theory correspondence,” *Phys. Rev. Lett.* **96**, 181602 (2006).

[37] V. Coffman, J. Kundu, and W. K. Wootters, “Distributed entanglement,” *Phys. Rev. A* **61**, 052306 (2000).

[38] W. H. Zurek, “Decoherence, einselection, and the quantum origins of the classical,” *Rev. Mod. Phys.* **75**, 715–775 (2003).

[39] G. Lindblad, “On the generators of quantum dynamical semigroups,” *Commun. Math. Phys.* **48**, 119–130 (1976).

[40] B. Skinner, J. Ruhman, and A. Nahum, “Measurement-induced phase transitions in the dynamics of entanglement,” *Phys. Rev. X* **9**, 031009 (2019).

[41] A. Chan, R. M. Nandkishore, M. Pretko, and G. Smith, “Unitary-projective entanglement dynamics,” *Phys. Rev. B* **99**, 224307 (2019).

[42] C. H. Bennett, G. Brassard, C. Crépeau, R. Jozsa, A. Peres, and W. K. Wootters, “Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels,” *Phys. Rev. Lett.* **70**, 1895–1899 (1993).

[43] M. Żukowski, A. Zeilinger, M. A. Horne, and A. K. Ekert, “‘Event-ready-detectors’ Bell experiment via entanglement swapping,” *Phys. Rev. Lett.* **71**, 4287–4290 (1993).

[44] C. H. Bennett and S. J. Wiesner, “Communication via one- and two-particle operators on Einstein-Podolsky-Rosen states,” *Phys. Rev. Lett.* **69**, 2881–2884 (1992).

[45] A. S. Holevo, “Bounds for the quantity of information transmitted by a quantum communication channel,” *Prob. Inf. Transm.* **9**, 177–183 (1973).

[46] P. W. Shor, “Scheme for reducing decoherence in quantum computer memory,” *Phys. Rev. A* **52**, R2493 (1995).

[47] A. M. Steane, “Error correcting codes in quantum theory,” *Phys. Rev. Lett.* **77**, 793–797 (1996).

[48] A. R. Calderbank and P. W. Shor, “Good quantum error-correcting codes exist,” *Phys. Rev. A* **54**, 1098–1105 (1996).

[49] E. Knill and R. Laflamme, “Theory of quantum error-correcting codes,” *Phys. Rev. A* **55**, 900–911 (1997).

[50] H. Reeh and S. Schlieder, “Bemerkungen zur Unitäräquivalenz von Lorentzinvarianten Feldern,” *Nuovo Cimento* **22**, 1051–1068 (1961).

[51] W. G. Unruh, “Notes on black-hole evaporation,” *Phys. Rev. D* **14**, 870–892 (1976).

[52] S. W. Hawking, “Breakdown of predictability in gravitational collapse,” *Phys. Rev. D* **14**, 2460–2473 (1976).

[53] S. W. Hawking, “Black hole explosions?” *Nature* **248**, 30–31 (1974).

[54] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* **43**, 199–220 (1975).

[55] D. N. Page, “Information in black hole radiation,” *Phys. Rev. Lett.* **71**, 3743–3746 (1993).

[56] P. Hayden and J. Preskill, “Black holes as mirrors: quantum information in random subsystems,” *J. High Energy Phys.* **2007**(09), 120 (2007).

[57] J. Maldacena and L. Susskind, “Cool horizons for entangled black holes,” *Fortschr. Phys.* **61**, 781–811 (2013).

[58] H. J. Kimble, “The quantum internet,” *Nature* **453**, 1023–1030 (2008).

[59] S. Wehner, D. Elkouss, and R. Hanson, “Quantum internet: A vision for the road ahead,” *Science* **362**, eaam9288 (2018).

[60] A. Pironio et al., “Random numbers certified by Bell’s theorem,” *Nature* **464**, 1021–1024 (2010).

[61] R. Colbeck, “Quantum and relativistic protocols for secure multi-party computation,” PhD thesis, University of Cambridge (2011); also arXiv:0911.3814 [quant-ph].

[62] M. Van Raamsdonk, "Building up spacetime with quantum entanglement," *Gen. Rel. Grav.* **42**, 2323–2329 (2010).

[63] R. Horodecki, P. Horodecki, M. Horodecki, and K. Horodecki, "Quantum entanglement," *Rev. Mod. Phys.* **81**, 865–942 (2009).

[64] W. K. Wootters and W. H. Zurek, "A single quantum cannot be cloned," *Nature* **299**, 802–803 (1982).

[65] A. K. Pati and S. L. Braunstein, "Impossibility of deleting an unknown quantum state," *Nature* **404**, 164–165 (2000).

[66] B. Schumacher and M. D. Westmoreland, "Sending classical information via noisy quantum channels," *Phys. Rev. A* **56**, 131–138 (1997).

[67] S. Aaronson, "Quantum computing since Democritus," (Cambridge University Press, Cambridge, 2013).

[68] A. Y. Kitaev, "Fault-tolerant quantum computation by anyons," *Ann. Phys.* **303**, 2–30 (2003).

[69] A. Y. Kitaev, "Quantum computations: algorithms and error correction," *Russ. Math. Surveys* **52**, 1191–1249 (1997).

[70] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, Cambridge, 2000).

[71] C. M. Dawson and M. A. Nielsen, "The Solovay-Kitaev algorithm," *Quantum Inf. Comput.* **6**, 81–95 (2006).

[72] R. Penrose, *The Emperor's New Mind* (Oxford University Press, Oxford, 1989).

[73] R. Penrose, *Shadows of the Mind* (Oxford University Press, Oxford, 1994).

[74] J. S. Bell, *Speakable and Unspeakable in Quantum Mechanics* (Cambridge University Press, Cambridge, 1987).

[75] J. Maldacena, "The large N limit of superconformal field theories and supergravity," *Int. J. Theor. Phys.* **38**, 1113–1133 (1999); also *Adv. Theor. Math. Phys.* **2**, 231–252 (1998).

[76] J. Maldacena, "Eternal black holes in anti-de Sitter," *J. High Energy Phys.* **2003**(04), 021 (2003).

[77] A. K. Ekert, “Quantum cryptography based on Bell’s theorem,” *Phys. Rev. Lett.* **67**, 661–663 (1991).

[78] C. H. Bennett and G. Brassard, “Quantum cryptography: public key distribution and coin tossing,” in *Proc. IEEE Int. Conf. on Computers, Systems and Signal Processing* (IEEE, New York, 1984), pp. 175–179.

[79] P. W. Shor, “Algorithms for quantum computation: discrete logarithms and factoring,” in *Proc. 35th Annual Symposium on Foundations of Computer Science* (IEEE, Los Alamitos, 1994), pp. 124–134.

[80] A. Zeilinger, “Experiment and the foundations of quantum physics,” *Rev. Mod. Phys.* **71**, S288–S297 (1999).

[81] A. Zeilinger, *Dance of the Photons: From Einstein to Quantum Teleportation* (Farrar, Straus and Giroux, New York, 2010).

[82] E. P. Wigner, “On the quantum correction for thermodynamic equilibrium,” *Phys. Rev.* **40**, 749–759 (1932).

[83] E. P. Wigner, “The unreasonable effectiveness of mathematics in the natural sciences,” *Commun. Pure Appl. Math.* **13**, 1–14 (1960).

[84] R. F. Werner, “Quantum states with Einstein-Podolsky-Rosen correlations admitting a hidden-variable model,” *Phys. Rev. A* **40**, 4277–4281 (1989).

[85] A. Connes and J. Roe, *Noncommutative Geometry* (Academic Press, San Diego, 1994).

[86] R. Penrose, “Angular momentum: an approach to combinatorial space-time,” in T. Bastin (ed.), *Quantum Theory and Beyond* (Cambridge University Press, Cambridge, 1971), pp. 151–180.

[87] C. Rovelli and L. Smolin, “Spin networks and quantum gravity,” *Phys. Rev. D* **52**, 5743–5759 (1995).

[88] E. Witten, “Topological quantum field theory,” *Commun. Math. Phys.* **117**, 353–386 (1988).

[89] E. Witten, “Quantum field theory and the Jones polynomial,” *Commun. Math. Phys.* **121**, 351–399 (1989).

[90] J. J. Bisognano and E. H. Wichmann, “On the duality condition for a Hermitian scalar field,” *J. Math. Phys.* **16**, 985–1007 (1975).

[91] R. Horodecki, M. Horodecki, and P. Horodecki, “Teleportation, Bell’s inequalities and inseparability,” *Phys. Lett. A* **222**, 21–25 (1996).

[92] P. Horodecki, “Separability criterion and inseparable mixed states with positive partial transposition,” *Phys. Lett. A* **232**, 333–339 (1997).

[93] M. Horodecki, P. Horodecki, and R. Horodecki, “Separability of mixed states: necessary and sufficient conditions,” *Phys. Lett. A* **223**, 1–8 (1996).

[94] O. Gühne and G. Tóth, “Entanglement detection,” *Phys. Rep.* **474**, 1–75 (2009).

[95] M. B. Plenio and S. Virmani, “An introduction to entanglement measures,” *Quantum Inf. Comput.* **7**, 1–51 (2007).

[96] C. E. Shannon, “A mathematical theory of communication,” *Bell Syst. Tech. J.* **27**, 379–423, 623–656 (1948).

[97] P. W. Shor and J. Preskill, “Simple proof of security of the BB84 quantum key distribution protocol,” *Phys. Rev. Lett.* **85**, 441–444 (2000).

[98] D. Gottesman, “Stabilizer codes and quantum error correction,” PhD thesis, California Institute of Technology (1997).

[99] P. Zanardi and M. Rasetti, “Noiseless quantum codes,” *Phys. Rev. Lett.* **79**, 3306–3309 (1997).

[100] M. Srednicki, “Entropy and area,” *Phys. Rev. Lett.* **71**, 666–669 (1993).

[101] C. Holzhey, F. Larsen, and F. Wilczek, “Geometric and renormalized entropy in conformal field theory,” *Nucl. Phys. B* **424**, 443–467 (1994).

[102] P. Calabrese and J. Cardy, “Entanglement entropy and quantum field theory,” *J. Stat. Mech.* **2004**, P06002 (2004).

[103] N. Lashkari, M. B. McDermott, and M. Van Raamsdonk, “Gravitational dynamics from entanglement ‘thermodynamics’,” *J. High Energy Phys.* **2014**(04), 195 (2014).

[104] T. Faulkner, M. Guica, T. Hartman, R. C. Myers, and M. Van Raamsdonk, “Gravitation from entanglement in holographic CFTs,” *J. High Energy Phys.* **2014**(03), 051 (2014).

[105] X.-L. Qi and D. Ranard, “Determining a local Hamiltonian from a single eigenstate,” *Quantum* **3**, 159 (2019).

[106] A. Almheiri, X. Dong, and D. Harlow, “Bulk locality and quantum error correction in AdS/CFT,” *J. High Energy Phys.* **2015**(04), 163 (2015).

[107] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, “Holographic quantum error-correcting codes: toy models for the bulk/boundary correspondence,” *J. High Energy Phys.* **2015**(06), 149 (2015).

[108] G. Vidal, “Entanglement renormalization,” *Phys. Rev. Lett.* **99**, 220405 (2007).

[109] B. Swingle, “Entanglement renormalization and holography,” *Phys. Rev. D* **86**, 065007 (2012).

[110] A. Almheiri, R. Mahajan, J. Maldacena, and Y. Zhao, “The Page curve of Hawking radiation from semiclassical geometry,” *J. High Energy Phys.* **2020**(03), 149 (2020).

[111] G. Penington, “Entanglement wedge reconstruction and the information paradox,” *J. High Energy Phys.* **2020**(09), 002 (2020).

[112] M. Hanks, “Spacetime Emergence in Phase Theory,” Independent Institute for Foundational Physics (2026). [Phase Theory Series, Paper 3].

[113] M. Hanks, “The Particle Spectrum from Phase Topology,” Independent Institute for Foundational Physics (2026). [Phase Theory Series, Paper 7].

[114] M. Hanks, “Gravity as Phase Curvature,” Independent Institute for Foundational Physics (2026). [Phase Theory Series, Paper 12].

[115] M. Hanks, “Phase Theory and the Standard Model,” Independent Institute for Foundational Physics (2026). [Phase Theory Series, Paper 18].

[116] M. Hanks, “Holography from Global Phase Constraints,” Independent Institute for Foundational Physics (2026). [Phase Theory Series, Paper 29].

Preprint submitted to the Phase Theory Series, Paper 30. Independent Institute for Foundational Physics. January 9, 2026.

All proofs, definitions, propositions, theorems, corollaries, and remarks are original unless otherwise noted.

Correspondence:

/cdn-cgi/l/email-protection